

Adaptive Super-Twisting Sliding Mode Control for Spiral Trajectory of a 3-DOF Robotic Manipulator in Automated Injera Batter Dispensing

Abstract

Injera, a staple Ethiopian flatbread, is traditionally prepared through manual batter dispensing onto a circular heated griddle, a process prone to inconsistency, hygiene concerns, and limited scalability in industrial settings. This paper presents the design, kinematic modeling, inward spiral trajectory planning, and robust control of a three-degree-of-freedom (3-DOF) SCARA robotic manipulator for automated injera batter dispensing. The kinematic model is derived using the Denavit-Hartenberg convention, and the manipulator's reachable workspace annulus is analytically verified to fully encompass the planned spiral dispensing path. An Archimedes spiral trajectory with quintic polynomial interpolation is formulated to achieve smooth, uniform batter distribution across the circular dispensing surface. To ensure robust tracking under dynamic uncertainties and external disturbances, an Adaptive Super-Twisting Sliding Mode Controller (ASTSMC) is designed, with online gain adaptation eliminating the requirement for prior knowledge of disturbance bounds. Lyapunov-based stability analysis guarantees finite-time convergence of the tracking error. Simulation results in MATLAB/Simulink demonstrate that the proposed ASTSMC significantly outperforms both a classical discontinuous sliding mode controller and a fixed-gain super-twisting controller in trajectory tracking accuracy, disturbance rejection, and chattering suppression, achieving a 34.5–40.7% reduction in ITAE relative to a fixed-gain super-twisting controller, and up to 63.0% relative to a classical sliding mode controller, while requiring only 1.4–2.9% additional control effort across all tested scenarios. The proposed framework establishes a foundation for intelligent automation of traditional food processing systems.

1. Introduction

Injera, a fermented sourdough flatbread traditionally prepared from teff flour, is the staple food of more than one hundred million people in Ethiopia and Eritrea [1]–[3]. During preparation, liquid batter is dispensed onto a heated circular baking plate (mitad) in a continuous spiral-like motion to produce a thin, uniformly distributed layer before steaming. As demand for injera continues to grow because of rapid urbanization, commercial-scale production, and expanding export markets, batter dispensing remains predominantly a manual operation [4]. Consequently, product quality depends heavily on the operator's skill, making batter distribution, layer thickness, and surface coverage difficult to reproduce consistently. Moreover, prolonged manual dispensing exposes workers to high temperatures and repetitive motions, limits productivity, and complicates hygienic operation in high-throughput production environments where repeated human interaction with the dispensing process is unavoidable [5]. These limitations motivate the development of an automated robotic system capable of dispensing injera batter accurately, consistently, and hygienically while preserving the traditional spreading pattern.

Robotic manipulation is well established in modern food processing, with applications such as pick-and-place operations, food packaging, cake decoration, food extrusion printing, and manipulation of viscous media such as batter and dough [6], [7]. These applications demonstrate how robotic manipulators can increase accuracy, repeatability, hygiene, and efficiency while minimizing manual involvement [8]. However, there appears to be limited work on the robotic automation of African food products, even though many studies have focused on Western and East Asian foods [8], [9]. To the authors' knowledge, there are no published studies describing an integrated robotic system for injera batter dispensing. Unlike general food handling, the robotic automation of injera batter dispensing requires coverage of a heated circular baking plate while maintaining the proper height between the nozzle and the baking plate. Because the batter is highly viscous and gravity-fed, trajectory planning and precise motion control are critical to achieving high-quality products. These requirements call for the integrated design of an appropriate manipulator architecture, an efficient coverage trajectory, and a robust nonlinear control strategy. Considering the predominantly planar nature of the dispensing task together with the need for independent regulation of the nozzle-to-surface distance, a three-degree-of-freedom R-R-P SCARA manipulator provides a suitable mechanical configuration for automated injera batter dispensing.

Trajectory planning plays a critical role in determining coverage quality and motion smoothness during robotic dispensing. Conventional raster or zigzag trajectories are widely used because of their simplicity but require repeated direction reversals that introduce abrupt accelerations and decelerations, potentially disturbing viscous materials and reducing deposition uniformity [10]. In contrast, continuous spiral trajectories have demonstrated improved process stability, coverage efficiency, and deposition quality in applications such as additive manufacturing, robotic coating, and material extrusion [11], [12]. However, their application to gravity-fed injera batter dispensing on a heated circular baking surface, particularly with coordinated vertical nozzle motion to maintain an appropriate dispensing height, has not been investigated. Accurate execution of such trajectories further requires a control strategy capable of handling the nonlinear dynamics and uncertainties of robotic manipulators [13].

Accurate execution of the proposed dispensing trajectory requires a control strategy capable of maintaining high tracking performance despite nonlinear coupled dynamics, friction, payload variation, and unmodeled disturbances. In the present application, these uncertainties are further compounded by the continuously changing batter mass in the dispensing container, which may degrade the performance of conventional linear controllers. Sliding Mode Control (SMC) is widely recognized for its robustness against matched uncertainties; however, classical SMC often suffers from chattering and relatively slow convergence, limiting tracking accuracy and actuator smoothness [14]. Super-twisting sliding mode control (STSMC) alleviates these drawbacks by providing continuous control action while preserving robustness [15], [16]. Furthermore, adaptive super-twisting formulations combined with terminal or nonsingular terminal sliding surfaces achieve finite-time convergence, reduce chattering, and improve robustness under uncertain operating conditions [17], [13], [18], [19].

In recent years, adaptive super-twisting and terminal sliding mode control approaches have been gaining popularity due to their ability to provide enhanced accuracy and robustness in trajectory tracking applications for robot manipulators. It has been proved that the application of adaptive gain laws, global or integral sliding surfaces, and higher-order sliding mode approaches provides increased robustness to disturbances, chatter suppression, and finite time convergence in the presence of uncertainties like friction, load changes, backlash, actuator failures, and parametric uncertainties [15], [17], [18], [20]–[28].

Despite these advances, existing studies primarily address generic manipulator trajectory-tracking problems rather than application-specific food dispensing tasks. Most investigations validate their controllers using standard benchmark trajectories, such as sinusoidal or exponential reference paths, on 2-DOF, 3-DOF, 6-DOF, or 7-DOF manipulators, while dispensing-specific requirements, including continuous spiral deposition, viscous material handling, hygienic operation, and uniform surface coverage, are rarely considered [14], [20], [21], [28]. Furthermore, manipulator design, trajectory planning, and robust nonlinear control have largely been investigated as separate problems. To the best of the authors' knowledge, no previous study has integrated these components into a unified robotic system for automated injera batter dispensing using a coordinated spiral trajectory and an adaptive higher-order sliding mode controller. Addressing this gap constitutes the primary objective of the present study.

To address these challenges, this paper proposes an integrated robotic system for automated injera batter dispensing by using an application-specific manipulator, coverage-oriented trajectory generation, and nonlinear robust control techniques in an integrated manner. A three-degree-of-freedom R-R-P SCARA manipulator is developed for the dispensing task, with its kinematic and dynamic models established to support accurate motion planning and controller design.

An inward Archimedean spiral trajectory synchronized with coordinated vertical nozzle motion is proposed to achieve continuous coverage of the circular baking surface while maintaining an appropriate dispensing height. To follow the generated trajectory, an adaptive super-twisting sliding mode controller is designed based on the authors' previous study [22] and classic super-twisting convergence theory. The adaptive gain regulation approach is used to improve the robustness of the controller in the presence of uncertain factors while minimizing the chattering effect and providing

finite-time convergence. The major contributions of this paper can be summarized as follows. The main contributions of this work are summarized as follows.

- Application-specific manipulator modeling: A three-degree-of-freedom R–R–P SCARA manipulator is developed for automated injera batter dispensing. Complete kinematic and dynamic models are established to provide the foundation for trajectory planning and robust controller design.
- Coverage-oriented spiral trajectory planning: An inward spiral dispensing trajectory is proposed to reproduce the traditional manual batter spreading process while ensuring continuous coverage of the circular baking surface. The trajectory is synchronized with the coordinated vertical motion of the prismatic joint to regulate the nozzle-to-surface distance throughout dispensing. A quintic time-scaling law is incorporated to generate smooth position, velocity, and acceleration profiles suitable for practical robotic implementation.
- Robust adaptive trajectory tracking: An Adaptive Super-Twisting Sliding Mode Controller (ASTSMC) is developed for accurate trajectory tracking under parametric uncertainties and external disturbances. Unlike conventional adaptive super-twisting approaches, the proposed controller employs a structurally coupled adaptive gain mechanism that preserves the gain relationship required for finite-time convergence while reducing chattering without requiring prior knowledge of disturbance bounds.
- Comprehensive performance evaluation: The effectiveness of the proposed control strategy is validated through comparative simulation studies against conventional PID and first-order Sliding Mode Control (SMC) under identical operating conditions. Comparative simulations demonstrate improved trajectory tracking accuracy, robustness, reduced control effort, and effective chattering suppression.

The remainder of this paper is organized as follows. Section II presents the mechanical configuration and analytical modeling of the proposed manipulator. Section III describes the spiral trajectory planning method. Section IV develops the Adaptive Super-Twisting Sliding Mode Controller and presents the stability analysis. Section V reports comparative simulation results, and Section VI concludes the paper and discusses future work.

2. Robot Design and Kinematic Modeling

The objective of this section is to formulate the geometrical structure of the proposed robotic manipulator along with its detailed kinematic model, velocity jacobians, and the desired trajectory planning strategy. This mathematical foundation ensures precise tracking of spatial trajectories during the automated dispensing process.

2.1 Mechanical Design

2.1.1 Link and Joint Configuration

The design of the manipulator is such that it forms a SCARA-type 3-degree-of-freedom (3-DOF) robotic arm using an R-R-P (Revolute-Revolute-Prismatic) Kinematic Chain to perform automated dispensing of the injera batter on the traditional circular Ethiopian baking surface referred to as mitad. The selection of this design stems from the fact that the movement of the first two revolute joints in a horizontal direction will perfectly mirror the flat nature of the mitad, allowing for accurate pour spiral tracking, while the final prismatic joint simultaneously retracts during the revolution. This synchronized vertical retraction dynamically adjusts the nozzle height during the pour to manage batter flow and layer thickness, leveraging the SCARA architecture's high vertical stiffness to prevent unwanted wobbling or deflection throughout the 3D dispensing path [29, 30].

The mechanical arrangement features:

- **Joint 1 (Shoulder):** A revolute joint located at the top of a rigid, floor-mounted vertical support column that provides the primary rotational sweep in the horizontal plane.
- **Joint 2 (Elbow):** A secondary revolute joint mounted at the distal end of Link 1, operating parallel to Joint 1 to achieve fine angular positioning.
- **Joint 3 (Vertical):** A vertical prismatic linear joint positioned at the distal end of Link 2 to translate the food-grade dispensing nozzle vertically above the cooking plate.

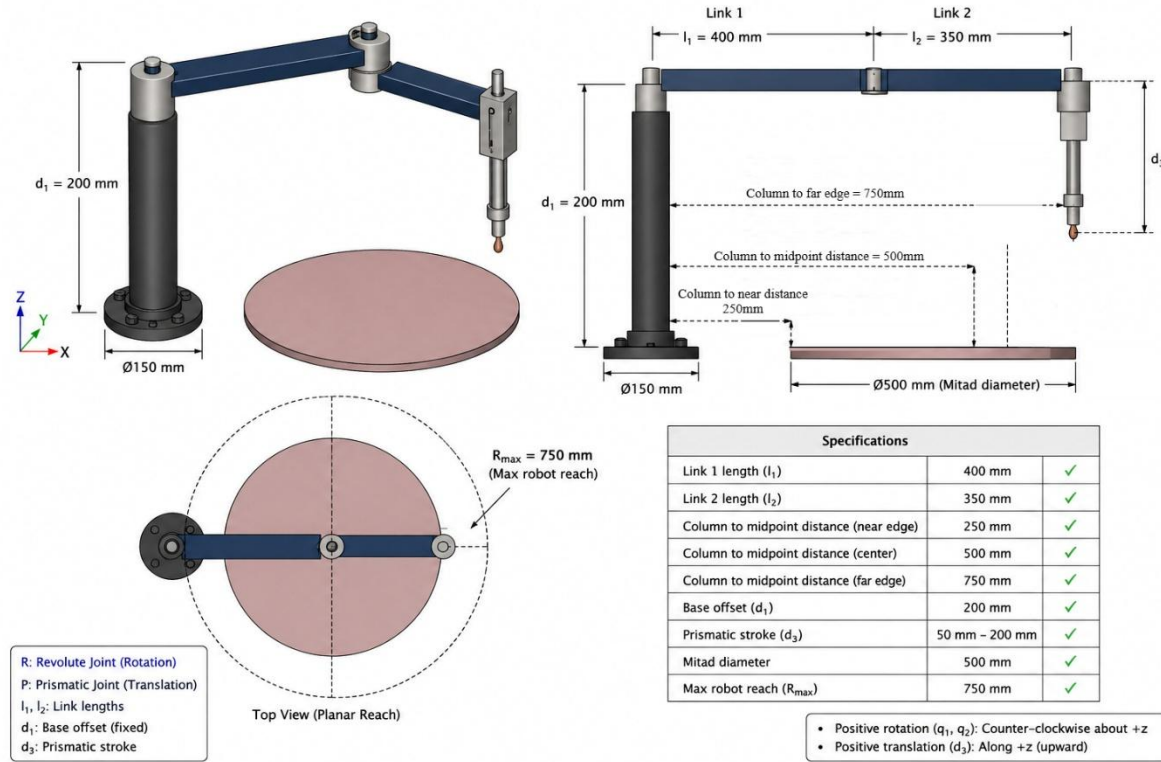


Figure 1: Mechanical architecture of the 3-DOF SCARA manipulator: (a) Isometric 3D view showing the R-R-P joint configuration relative to the mitad; (b) Top-down schematic illustrating the link lengths (l_1, l_2) and the annular workspace coverage.

2.1.3 Denavit-Hartenberg Parameters

The kinematic structure of the proposed 3-DOF SCARA manipulator is characterized using the Craig (Modified) Denavit-Hartenberg (DH) convention [31]. In this convention, four parameters are associated with each joint-link pair: the link length a_{i-1} , the link twist α_{i-1} , the link offset d_i , and the joint angle θ_i , as defined below.

a_{i-1}	Link length: distance from Z_{i-1} to Z_i measured along X_{i-1}
α_{i-1}	Link twist: angle from Z_{i-1} to Z_i about X_{i-1}
d_i	Link offset: distance from X_{i-1} to X_i along Z_i
θ_i	Joint angle: angle from X_{i-1} to X_i about Z_i

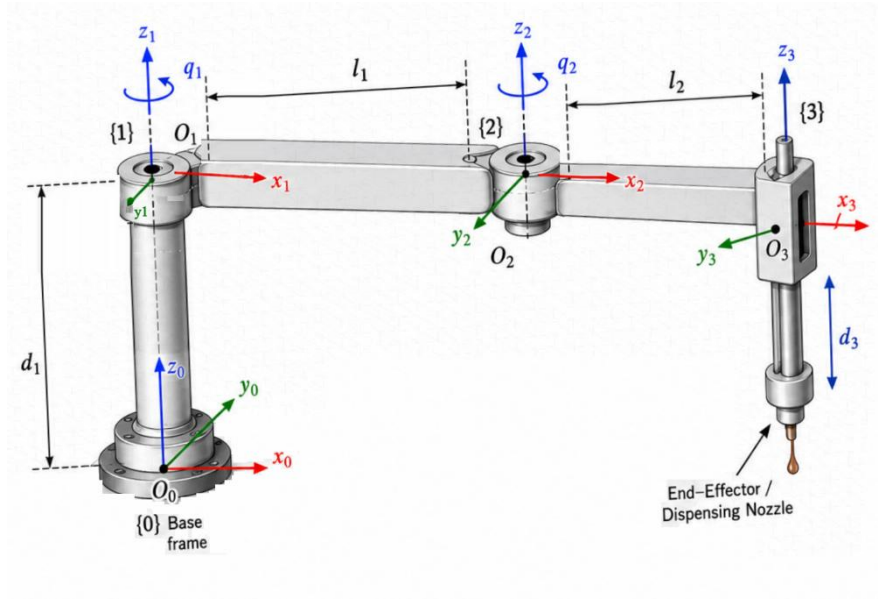


Figure 2: Coordinate frame assignments and joint variables for the R-R-P kinematic chain according to Craig's Modified Denavit-Hartenberg convention.

Following systematic frame assignment to each joint axis, the complete DH parameter table for the R–R–P SCARA manipulator is presented in Table 1. Because all three joint axes are parallel and vertical, all link twist angles satisfy $\alpha_{i-1} = 0^\circ$ throughout the kinematic chain.

Table 1. Modified DH Parameters for the 3-DOF SCARA Injera Dispensing Manipulator. (*) denotes the variable joint parameter.

Joint i	a_{i-1} (mm)	α_{i-1} (deg)	d_i (mm)	θ_i (deg)	Type
1 (Shoulder)	0	0	d_1 (fixed)	$\theta_1 *$	Revolute
2 (Elbow)	11	0	0	$\theta_2 *$	Revolute
3 (Vertical)	12	0	$d_3 *$	0	Prismatic

2.2 Forward Kinematics

2.2.1 Link Transformation Matrices

Substitution of the geometrical parameters shown in Table 2.1 in the homogeneous transformations leads to the transformation matrices of each link of the proposed SCARA robot. The transformation matrix between the base frame (0) and the frame (1), which corresponds to the shoulder joint, is expressed as

$${}^0T_1 = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 & 0 & 0 \\ \sin \theta_1 & \cos \theta_1 & 0 & 0 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The transformation from Frame (1) to Frame (2), representing the elbow joint, is expressed as

$${}^1_2T = \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 & 0 & l1 \\ \sin \theta_2 & \cos \theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Similarly, the transformation from Frame (2) to Frame (3), associated with the prismatic actuator, is defined by

$${}^2_3T = \begin{bmatrix} 1 & 0 & 0 & l2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

2.2.2 End-Effector Position and Orientation

The forward kinematic model of the manipulator is obtained by sequentially multiplying the individual homogeneous transformation matrices. Accordingly, the overall transformation relating the end-effector frame to the base frame is given by

$${}^0_3T = {}^0_1T {}^1_2T {}^2_3T.$$

The intermediate transformation matrix between the base frame and Frame (2) is obtained as

$${}^0_2T = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & l1\cos \theta_1 \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & l1\sin \theta_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Subsequent multiplication of 0_2T and 2_3T yields the complete forward kinematic model

$${}^0_3T = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & l1\cos \theta_1 + l2\cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & l1\sin \theta_1 + l2\sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & d_1 + d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The translational component of the resulting transformation matrix specifies the Cartesian position of the dispensing nozzle relative to the base coordinate system. Therefore, the end-effector position coordinates are expressed as

$$\begin{aligned} X &= l1\cos \theta_1 + l2\cos(\theta_1 + \theta_2), \\ Y &= l1\sin \theta_1 + l2\sin(\theta_1 + \theta_2), \\ Z &= d_1 + d_3. \end{aligned}$$

However, the rotational part of the transformation matrix shows that the orientation of the end-effector in the vertical direction depends on the total rotation $(\theta_1 + \theta_2)$ of the two revolute joints. Besides, there is an evident decoupling in the kinematic chain between the planar motion and the vertical motion of the robot. In this case, the planar coordinates of the nozzle are achieved through the use of the parameters of the revolute joints $(\theta_1 + \theta_2)$, while the vertical movement is separately achieved through the prismatic displacement d_3 .

2.3 Inverse Kinematics

The forward kinematics derived in Section 2.2 express the end-effector position (X, Y, Z) as an explicit function of the joint variables $(\theta_1, \theta_2, d_3)$. On the other hand, trajectory generation provides the desired path in task space and therefore requires the inverse map: given a target position (X, Y, Z) , determine the corresponding joint variables. Since the manipulator consists of a planar 2R sub-chain (θ_1, θ_2) with a separate prismatic axis (d_3), the inverse kinematics calculation reduces to the typical two-link planar subproblem and one linear equation.

Squaring and summing the planar forward-kinematic expressions for X and Y and applying the law of cosines to the triangle formed by l_1, l_2 and the radial distance to the target gives the cosine of the elbow angle,

The formulas shown in the image are:

$$\cos \theta_2 = \frac{X^2 + Y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

from which the elbow angle is recovered using the two-argument arctangent, preserving both the elbow-up and elbow-down configurations,

$$\theta_2 = \text{atan2} \left(\pm \sqrt{1 - \cos^2 \theta_2}, \cos \theta_2 \right)$$

where the $+$ sign corresponds to the elbow-down configuration and the $-$ sign to the elbow-up configuration. With θ_2 determined, the shoulder angle follows from the geometric decomposition of the planar position vector into the direction toward the target and the internal triangle angle,

where the $+$ sign corresponds to the elbow-down configuration and the $-$ sign corresponds to the elbow-up configuration.

$$\theta_1 = \text{atan2}(Y, X) - \text{atan2}(l_2 \sin \theta_2, l_1 + l_2 \cos \theta_2)$$

Finally, since the prismatic joint contributes only a vertical translation, the prismatic extension is obtained directly from the linear Z -equation of Section 2.2,

$$d_3 = Z - d_1$$

Equations (2.22)–(2.25) constitute the complete closed-form inverse kinematic solution of the manipulator. A target position is reachable by the planar sub-chain if and only if it lies within the annular workspace bounded by the fully folded and fully extended arm configurations,

$$|l_1 - l_2| \leq \sqrt{X^2 + Y^2} \leq l_1 + l_2$$

a condition that is verified for every target point generated by the spiral path of Section 2.5 prior to trajectory execution.

2.4 Jacobian and Velocity Kinematics

Having established the forward kinematics of the R–R–P SCARA manipulator in [Section 2.2](#), the next step is to relate the joint-space velocities $\dot{q}$ to the linear and angular velocities of each link. This mapping is captured by the *manipulator Jacobian*, which underlies velocity analysis, singularity assessment, static force transmission, and the kinetic-energy formulation needed to derive the dynamic model in [Section 3](#).

The generalized coordinate vector of the manipulator is

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}.$$

where θ_1 and θ_2 are the shoulder and elbow joint angles (revolute) and d_3 is the prismatic extension of the vertical dispensing axis.

2.3.1 Center-of-Mass Position Vectors

Since the inertial matrix depends on the velocity of the center of mass of each link, the first thing that we will do is determine the position of the COM of all links in the base frame, assuming that Links 1 and 2 are uniform slender bodies such that the COM of each body coincides with the geometric center of their lengths, and the sliding mass of the prismatic stage (Link 3) is concentrated at the slider.

Link 1 center of mass

$$p_{c1} = \begin{bmatrix} \frac{l_1}{2} \cos \theta_1 \\ \frac{l_1}{2} \sin \theta_1 \\ d_1 \end{bmatrix}$$

Link 2 center of mass

The origin of Joint 2 is

$$\begin{bmatrix} l_1 \cos \theta_1 \\ l_1 \sin \theta_1 \\ d_1 \end{bmatrix}$$

Hence, the COM is another $\frac{l_2}{2}$ along the direction $\theta_1 + \theta_2$.

$$p_{c2} = \begin{bmatrix} l_1 \cos \theta_1 + \frac{l_2}{2} \cos(\theta_1 + \theta_2) \\ l_1 \sin \theta_1 + \frac{l_2}{2} \sin(\theta_1 + \theta_2) \\ d_1 \end{bmatrix}$$

Link 3 (prismatic slider)

The slider moves vertically. Its COM is

$$p_{c3} = \begin{bmatrix} l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) \\ l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) \\ d_1 + d_3 \end{bmatrix}$$

Here, d_3 is the signed extension along the local Z-axis defined by the Modified DH convention adopted in Section 2.2. If the physical actuator drives the nozzle downward to increase extension, this is already accounted for by the sign convention embedded in the forward-kinematic expression $Z = d_1 + d_3$ and is carried forward consistently into the Jacobian below.

2.3.2 Linear Velocity Jacobian J_p

The linear Jacobian of link i maps the joint velocity vector $\dot{q}$ onto the linear velocity of that link's center of mass:

$$J_P = \frac{\partial p_c}{\partial q}$$

Each column of J_{P_i} is obtained by differentiating p_{ci} with respect to one generalized coordinate. Differentiating p_{c1} , p_{c2} , and p_{c3} derived above, column by column, yields the three 3×3 linear Jacobian matrices below.

Linear Jacobian of Link 1

Link 1 depends only on θ_1 , so the second and third columns vanish:

$$J_{P1} = \begin{bmatrix} -\frac{l_1}{2} \sin \theta_1 & 0 & 0 \\ \frac{l_1}{2} \cos \theta_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Linear Jacobian of Link 2

Link 2 depends on both θ_1 and θ_2 , but not on the prismatic coordinate d_3 :

$$J_{P2} = \begin{bmatrix} -l_1 \sin \theta_1 - \frac{l_2}{2} \sin(\theta_1 + \theta_2) & -\frac{l_2}{2} \sin(\theta_1 + \theta_2) & 0 \\ l_1 \cos \theta_1 + \frac{l_2}{2} \cos(\theta_1 + \theta_2) & \frac{l_2}{2} \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Linear Jacobian of Link 3 (dispensing nozzle)

Link 3 depends on all three joint variables. Its third column is $[0, 0, 1]^T$, reflecting that the prismatic joint produces a pure translation along the local Z-axis defined in the forward kinematics:

$$J_{P3} = \begin{bmatrix} -l_1 \sin \theta_1 - l_2 \sin(\theta_1 + \theta_2) & -l_2 \sin(\theta_1 + \theta_2) & 0 \\ l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) & l_2 \cos(\theta_1 + \theta_2) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.3.3 Angular Velocity Jacobian J_ω

The angular Jacobian follows the standard geometric rule for serial manipulators: a revolute-joint column contributes the unit vector of that joint's rotation axis, z_{i-1} , while a prismatic-joint column contributes zero, since a prismatic joint produces no rotation: [32]

$$j_{O_i} = z_{i-1} \text{ (revolute joint),} \quad j_{O_i} = 0 \text{ (prismatic joint)}$$

All three joint axes of the R–R–P SCARA structure are parallel to the base Z-axis ($z_0 = z_1 = z_2 = [0, 0, 1]^T$), since the manipulator rotates in a horizontal plane and translates vertically. Applying the rule above to each link gives:

$$J_{O1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$J_{O2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$J_{O3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

The zero third column in J_{O2} and J_{O3} confirms that the prismatic joint contributes no angular velocity to Link 2 or Link 3, a structural property of SCARA-type kinematics that substantially simplifies the inertia matrix derived next.

With J_{Pi} and J_{Oi} now available for all three links, every kinematic quantity needed to assemble the kinetic energy of the manipulator

2.5 Trajectory Planning

After developing the equations for forward and inverse kinematics, the dispensing trajectory to be followed is defined in the task space and then mapped to joint space using inverse kinematics. To mimic the conventional pouring process of injera, an inward Archimedes' spiral trajectory is used as the reference trajectory in task space.

2.5.1 Task-Space Spiral Path Generation

Unlike the conventional outward spiral, the proposed trajectory begins near the perimeter of the baking pan and progressively moves toward the center while the prismatic joint simultaneously retracts, thereby reducing the vertical extension of the dispensing mechanism during the dispensing process.

The spiral radius is defined by

$$r(\phi) = R - b\phi, \quad (2.27)$$

where R is the effective radius of the baking pan, b is the constant spiral pitch, and ϕ is the spiral angle introduced to distinguish it from the joint variables θ_1 and θ_2 .

The desired Cartesian trajectory of the dispensing nozzle is therefore expressed as

$$\mathbf{p}_d(\phi) = \begin{bmatrix} x_c + (R - b\phi)\cos\phi \\ y_c + (R - b\phi)\sin\phi \\ d_1 + d_3(\phi) \end{bmatrix}, \quad (2.28)$$

Where $\mathbf{p}_d(\phi)$ denotes the desired Cartesian position of the dispensing nozzle, (x_c, y_c) the center of the baking pan and d_1 is the fixed height of the shoulder joint.

To coordinate the vertical motion with the inward spiral, the prismatic displacement is prescribed as a linear function of the spiral parameter,

$$d_3(\phi) = d_{3,\max} - \frac{d_{3,\max} - d_{3,\min}}{\phi_{\max}} \phi, \quad (2.29)$$

where $d_{3,\max}$ and $d_{3,\min}$ denote the initial and final prismatic displacements, respectively, and $\phi_{\max}$ is the total spiral angle required to reach the center of the baking pan. Consequently, the prismatic joint retracts continuously as the dispensing nozzle progresses toward the center, resulting in a synchronized three-dimensional dispensing trajectory.

The proposed trajectory was chosen since it simulates the conventional way of preparing injera and ensures uniform coverage of the round baking surface. The suggested trajectory differs from raster or zigzag trajectory in that it does not involve abrupt direction changes and repeated acceleration–

deceleration cycles; therefore, it provides better motion control, dispensing stability, and batter dispersion.

2.5.2 Joint Trajectory Generation

The task-space trajectory defined by (2.28) is transformed into joint-space commands through the inverse kinematic model developed in Section 2.4. For each point along the spiral trajectory, the corresponding joint variables $\theta_1(\phi)$, $\theta_2(\phi)$, and $d_3(\phi)$ are obtained by substituting the desired Cartesian coordinates $X(\phi)$, $Y(\phi)$, and $Z(\phi)$ into the inverse kinematic equations (2.22)–(2.25). Consequently, the complete geometric trajectory is determined in the joint space before introducing the time parameterization.

To ensure smooth manipulator motion, a quintic polynomial is employed as the trajectory time law. Instead of assigning independent timing to each joint, the spiral parameter ϕ is expressed as a function of time,

$$\phi(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5, \quad (2.29)$$

where the coefficients $a_0, \dots, a_5$ are determined from the prescribed boundary conditions on position, velocity, and acceleration at the initial and final instants of the motion.

Differentiating (2.29) yields the spiral angular velocity and acceleration,

$$\dot{\phi}(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4, \quad (2.30)$$

$$\ddot{\phi}(t) = 2a_2 + 6a_3t + 12a_4t^2 + 20a_5t^3. \quad (2.31)$$

The resulting joint trajectories are obtained by composing the inverse kinematic solution with the time law,

$$\theta_1(t) = \theta_1(\phi(t)), \theta_2(t) = \theta_2(\phi(t)), d_3(t) = d_3(\phi(t)). \quad (2.32)$$

The quintic polynomial guarantees continuous position, velocity, and acceleration throughout the motion, providing smooth trajectory tracking while minimizing abrupt actuator commands and mechanical vibration. These characteristics are particularly desirable for automated injera dispensing, where continuous nozzle motion contributes to stable batter flow and uniform layer formation.

3. Dynamic Modeling

This section derives the complete dynamic model of the 3-DOF R–R–P SCARA manipulator directly from the kinematic quantities obtained in Section 2.3, following the Euler–Lagrange energy-based workflow: *Kinematics* $\rightarrow$ *Center of Mass* $\rightarrow$ *Jacobians* $\rightarrow$ *Kinetic Energy* $\rightarrow$ *Potential Energy* $\rightarrow$ *Lagrangian* $\rightarrow$ *Equations of Motion*. The resulting inertia matrix $B(q)$, Coriolis/centrifugal matrix $C(q, \dot{q})$, and gravity vector $G(q)$ form the model-based foundation on which the manipulator’s sliding-mode (ASTSMC) controller is designed in the following chapter.

3.1 Kinetic Energy and the Manipulator Inertia Matrix $B(q)$

Let m_1 , m_2 , and m_3 denote the masses of Link 1, Link 2, and the prismatic slider (including the nozzle/carriage assembly), respectively. Likewise, let I_1 , I_2 , and I_3 denote the corresponding moments of inertia about each link’s center-of-mass **Z-axis**. Since all joint axes of the SCARA manipulator are parallel to the base Z-axis, the angular velocity of each link is purely a rotation about Z. Consequently, the kinetic energy of the i -th link can be expressed as

$$T_i = \frac{1}{2} m_i \dot{p}_{c_i}^T \dot{p}_{c_i} + \frac{1}{2} I_i \omega_i^T \omega_i \quad (3.1)$$

Using the Jacobian relations derived in Section 2.3,

$$\dot{\mathbf{p}}_{c_i} = J_{P_i}(q)\dot{q}, \omega_i = J_{O_i}(q)\dot{q} \quad (3.2)$$

the kinetic energy becomes

$$T_i = \frac{1}{2} \dot{q}^T (m_i J_{P_i}^T J_{P_i} + J_{O_i}^T I_i J_{O_i}) \dot{q} \quad (3.3)$$

Summing the kinetic energies of all links yields

$$T = \sum_{i=1}^3 T_i = \frac{1}{2} \dot{q}^T B(q) \dot{q} \quad (3.4)$$

where the symmetric, positive-definite inertia matrix $B(q)$ is assembled directly from the Jacobians of Section 2.3:

$$B(q) = \sum_{i=1}^3 (m_i J_{P_i}^T J_{P_i} + J_{O_i}^T I_i J_{O_i}) \quad (3.5)$$

Substituting J_{P1} , J_{P2} , J_{P3} and J_{O1} , J_{O2} , J_{O3} from equations (2.16)–(2.21) and carrying out the matrix products term by term gives the four independent non-zero inertia coefficients:

$$B_{11} = I_1 + I_2 + I_3 + \frac{m_1 l_1^2}{4} + m_2 \left(l_1^2 + \frac{l_2^2}{4} + l_1 l_2 \cos \theta_2 \right) + m_3 (l_1^2 + l_2^2 + 2l_1 l_2 \cos \theta_2) \quad (3.6)$$

$$B_{12} = B_{21} = I_2 + I_3 + m_2 \left(\frac{l_2^2}{4} + \frac{1}{2} l_1 l_2 \cos \theta_2 \right) + m_3 (l_2^2 + l_1 l_2 \cos \theta_2) \quad (3.7)$$

$$B_{22} = I_2 + I_3 + \frac{m_2 l_2^2}{4} + m_3 l_2^2 \quad (3.8)$$

$$B_{33} = m_3 \quad (3.9)$$

Therefore, the complete inertia matrix of the 3-DOF R–R–P SCARA manipulator is

$$B(q) = \begin{bmatrix} B_{11} & B_{12} & 0 \\ B_{12} & B_{22} & 0 \\ 0 & 0 & m_3 \end{bmatrix} \quad (3.10)$$

Two important structural observations can immediately be made:

1. The prismatic coordinate d_3 is completely decoupled from the revolute coordinates θ_1 and θ_2 , since $B_{13} = B_{23} = 0$ and $B_{33} = m_3$ is constant.
2. The planar inertia terms B_{11} , B_{12} , and B_{22} depend only on θ_2 through the coupling term $\cos \theta_2$, which is identical to the classical two-link planar manipulator structure. The prismatic stage therefore behaves as an additional end mass attached to the planar 2R subsystem.

3.2 Potential Energy and Gravity Vector $G(q)$

With the base-frame Z-axis defined vertically upward and gravitational acceleration g acting downward, the total gravitational potential energy of the manipulator is obtained by summing the potential energy of all moving links:

$$V = m_1 g z_{c1} + m_2 g z_{c2} + m_3 g z_{c3} \quad (3.11)$$

From the center-of-mass positions derived in Section 2.3, the first two links move entirely within the horizontal XY -plane. Therefore, their center-of-mass heights remain constant and contribute only constant terms to the potential energy. The only coordinate-dependent height is associated with the prismatic slider:

$$z_{c3} = d_1 + d_3 \quad (3.12)$$

Substituting into (3.11) gives

$$V = g(m_1 z_{c1} + m_2 z_{c2} + m_3(d_1 + d_3)) \quad (3.13)$$

Since z_{c1} , z_{c2} , and the vertical offset d_1 are constants, the potential energy can be rewritten as

$$V = V_0 + m_3 g d_3 \quad (3.14)$$

where V_0 represents the constant portion of the gravitational energy. Because constant terms do not affect the equations of motion, only the coordinate-dependent term is retained in the Lagrangian formulation.

The gravity vector is obtained from the standard definition

$$G(q) = \frac{\partial V}{\partial q} = \left[\frac{\partial V}{\partial \theta_1}, \frac{\partial V}{\partial \theta_2}, \frac{\partial V}{\partial d_3} \right]^T \quad (3.15)$$

Evaluating the partial derivatives yields

$$\frac{\partial V}{\partial \theta_1} = 0, \frac{\partial V}{\partial \theta_2} = 0, \frac{\partial V}{\partial d_3} = m_3 g \quad (3.16)$$

and therefore the gravity vector becomes

$$G(q) = \begin{bmatrix} 0 \\ 0 \\ m_3 g \end{bmatrix} \quad (3.17)$$

This characteristic is particularly advantageous for the injera-dispensing application because the planar positioning subsystem (θ_1, θ_2) is completely gravity-decoupled, while the vertical actuator must compensate only for the constant load $m_3 g$. Such a structure simplifies both controller design and stability analysis in the subsequent ASTSMC development.

3.3 Lagrangian Formulation

Having established the total kinetic energy $T(q, \dot{q})$ in Section 3.1 and the total potential energy $V(q)$ in Section 3.2, the dynamic equations of the proposed 3-DOF R–R–P SCARA manipulator are derived using the Euler–Lagrange formulation. The Lagrangian function is defined as

$$L(q, \dot{q}) = T(q, \dot{q}) - V(q)$$

where the generalized coordinate vector and generalized velocity vector are

$$q = [\theta_1 \quad \theta_2 \quad d_3]^T, \dot{q} = [\dot{\theta}_1 \quad \dot{\theta}_2 \quad \dot{d}_3]^T.$$

Applying the Euler–Lagrange equation to each generalized coordinate yields

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \tau_i, i = 1, 2, 3$$

where τ_i denotes the generalized force associated with the i -th coordinate.

Since the kinetic energy has already been expressed in the quadratic form

$$T = \frac{1}{2} \dot{q}^T B(q) \dot{q},$$

the derivative of the kinetic energy with respect to the generalized velocity becomes

$$\frac{\partial T}{\partial \dot{q}} = B(q) \dot{q}.$$

Taking the time derivative gives

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}} \right) = \frac{d}{dt} (B(q) \dot{q}) = B(q) \ddot{q} + \dot{B}(q) \dot{q},$$

where $B(q) \ddot{q}$ represents the inertial acceleration term and $\dot{B}(q) \dot{q}$ arises from the configuration dependence of the inertia matrix.

Similarly,

$$\frac{\partial L}{\partial q} = \frac{\partial T}{\partial q} - \frac{\partial V}{\partial q}.$$

From Section 3.2,

$$G(q) = \frac{\partial V}{\partial q} = \begin{bmatrix} 0 \\ 0 \\ m_3 g \end{bmatrix},$$

and therefore

$$\frac{\partial L}{\partial q} = \frac{\partial T}{\partial q} - G(q).$$

Substituting these expressions into the Euler–Lagrange equation yields

$$B(q) \ddot{q} + \dot{B}(q) \dot{q} - \frac{\partial T}{\partial q} + G(q) = \tau.$$

The quantity

$$\dot{B}(q) \dot{q} - \frac{\partial T}{\partial q}$$

contains all velocity-dependent coupling effects of the manipulator. These terms are conventionally grouped into the Coriolis and centrifugal contribution of the dynamic model.

To obtain a compact matrix representation, the Christoffel symbols of the first kind are introduced as

$$c_{ijk} = \frac{1}{2} \left(\frac{\partial B_{ij}}{\partial q_k} + \frac{\partial B_{ik}}{\partial q_j} - \frac{\partial B_{jk}}{\partial q_i} \right).$$

The Coriolis and centrifugal matrix is then assembled according to

$$C_{ij}(q, \dot{q}) = \sum_{k=1}^3 c_{ijk} \dot{q}_k.$$

Because the inertia matrix derived in Section 3.1 depends only on the elbow angle θ_2 , the only non-zero partial derivatives are

$$\frac{\partial B_{11}}{\partial \theta_2}, \frac{\partial B_{12}}{\partial \theta_2} = \frac{\partial B_{21}}{\partial \theta_2}.$$

Defining the coupling coefficient

$$h = \left(\frac{m_2}{2} + m_3 \right) l_1 l_2 \sin \theta_2,$$

the resulting Coriolis and centrifugal matrix becomes

$$C(q, \dot{q}) = \begin{bmatrix} -h\dot{\theta}_2 & -h(\dot{\theta}_1 + \dot{\theta}_2) & 0 \\ h\dot{\theta}_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Consequently, the velocity-dependent term may be written compactly as

$$\dot{B}(q)\dot{q} - \frac{\partial T}{\partial q} = C(q, \dot{q})\dot{q}.$$

The equation above represents the complete Coriolis and centrifugal effects of the manipulator. As expected from the structure of the inertia matrix, the third row and third column are identically zero, confirming that the vertical prismatic dispensing axis is dynamically decoupled from the planar revolute subsystem. This property significantly simplifies both the dynamic analysis and the controller design presented in the following chapter.

3.5 Complete Equations of Motion

Substituting the inertia matrix $B(q)$ derived in Section 3.1, the gravity vector $G(q)$ derived in Section 3.2, and the Coriolis/centrifugal matrix $C(q, \dot{q})$ obtained in Section 3.3 into the dynamic model yields the complete equations of motion of the 3-DOF R–R–P SCARA manipulator:

$$B(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau \quad (3.33)$$

where

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ d_3 \end{bmatrix}, \tau = \begin{bmatrix} \tau_1 \\ \tau_2 \\ F_3 \end{bmatrix}. \quad (3.34)$$

Here, τ_1 and τ_2 denote the actuator torques applied at the shoulder and elbow joints, respectively, while F_3 represents the actuation force of the prismatic dispensing axis.

Substituting the explicit expressions for $B(q)$, $C(q, \dot{q})$, and $G(q)$ gives

$$\begin{bmatrix} B_{11} & B_{12} & 0 \\ B_{12} & B_{22} & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{d}_3 \end{bmatrix} + \begin{bmatrix} -h\dot{\theta}_2 & -h(\dot{\theta}_1 + \dot{\theta}_2) & 0 \\ h\dot{\theta}_1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ m_3 g \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \\ F_3 \end{bmatrix}. \quad (3.35)$$

Expanding the matrix equation yields the individual joint equations of motion:

For Joint 1,

$$B_{11}\ddot{\theta}_1 + B_{12}\ddot{\theta}_2 - h(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) = \tau_1 \quad (3.36)$$

For Joint 2,

$$B_{12}\ddot{\theta}_1 + B_{22}\ddot{\theta}_2 + h\dot{\theta}_1^2 = \tau_2 \quad (3.37)$$

For the prismatic dispensing axis,

$$m_3\ddot{d}_3 + m_3g = F_3. \quad (3.38)$$

Equations (3.36)–(3.38) constitute the complete rigid-body dynamic model of the proposed SCARA injera-dispensing manipulator. These equations fully describe the inertial, velocity-coupling, and gravitational effects acting on the system and provide the model-based foundation for the Adaptive Super-Twisting Sliding Mode Controller (ASTSMC) developed in the next chapter.

The structure of the model reveals that the planar revolute subsystem (θ_1, θ_2) exhibits the classical nonlinear coupling of a two-link robotic arm, whereas the vertical prismatic axis remains dynamically decoupled except for the constant gravitational load m_3g . This property significantly simplifies controller design and facilitates

4. Adaptive Super-Twisting Sliding Mode Control

This chapter develops the adaptive super-twisting sliding-mode controller (ASTSMC) used to track the inward-spiral dispensing trajectory of Section 2.5 on the dynamic model identified in Section 3. The design proceeds in four stages: a sliding-surface definition that reduces the second-order tracking problem to a first-order reachability problem (Section 4.2); a model-based equivalent-control term that cancels the known nonlinear dynamics (Section 4.3); a super-twisting switching law that rejects the remaining matched disturbance without the high-frequency chattering of first-order sliding-mode control (Section 4.4); and an adaptive gain law that removes the need for prior knowledge of the disturbance bound (Section 4.5). Section 4.7 summarizes the finite-time convergence guarantee, citing the established Lyapunov results on which it rests.

4.1 Control Objective and Problem Statement

Let

$$\mathbf{q}(t) = \begin{bmatrix} \theta_1(t) \\ \theta_2(t) \\ d_3(t) \end{bmatrix} \in \mathbb{R}^3$$

denote the generalized coordinates of the 3-DOF RRP injera-dispensing manipulator, and let

$$\mathbf{q}_d(t) = \begin{bmatrix} \theta_{1d}(t) \\ \theta_{2d}(t) \\ d_{3d}(t) \end{bmatrix}$$

be the desired joint trajectory obtained from the inverse kinematics of the spiral dispensing path described in Sections 2.3 and 2.5.

The manipulator dynamics identified in Section 3 are represented by

$$\mathbf{B}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \boldsymbol{\tau} + \boldsymbol{\Delta}(\mathbf{q}, \dot{\mathbf{q}}, t), \quad (4.1)$$

where

- $\mathbf{B}(\mathbf{q}) \in \mathbb{R}^{3 \times 3}$ is the symmetric positive-definite inertia matrix,
- $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{3 \times 3}$ denotes the Coriolis and centrifugal effects,
- $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^3$ represents the gravitational vector,
- $\boldsymbol{\tau} \in \mathbb{R}^3$ is the control input,
- $\boldsymbol{\Delta}(\mathbf{q}, \dot{\mathbf{q}}, t) \in \mathbb{R}^3$ is a lumped matched disturbance incorporating unmodeled dynamics, friction, structural flexibility, and variations in batter viscosity during the dispensing process.

Assumption 1. The disturbance vector $\boldsymbol{\Delta}(t)$ is continuously differentiable and possesses a bounded derivative. That is, there exists an unknown constant $L > 0$ such that

$$\|\dot{\boldsymbol{\Delta}}(t)\| \leq L, \quad (4.2)$$

for all admissible trajectories of the manipulator. This assumption is physically reasonable for the dispensing application because variations in batter viscosity, friction, and payload occur continuously and therefore induce disturbances with bounded rates of change.[\[33\]](#)

Define the tracking error as

$$\mathbf{e}(t) = \mathbf{q}_d(t) - \mathbf{q}(t). \quad (4.3)$$

The control objective is to design a robust control law $\boldsymbol{\tau}$ such that the tracking error, $\mathbf{e}(t)$, converges in finite time to an arbitrarily small neighborhood of the origin, on a designed sliding manifold, despite unknown matched disturbance $\boldsymbol{\Delta}$, and without requiring prior knowledge of their bounds L .

4.2 Sliding Surface Design

Define the joint-space tracking error and its time derivative as

$$\mathbf{e}(t) = \mathbf{q}_d(t) - \mathbf{q}(t), \quad (4.4)$$

and

$$\dot{\mathbf{e}}(t) = \dot{\mathbf{q}}_d(t) - \dot{\mathbf{q}}(t). \quad (4.5)$$

A first-order sliding surface is selected as

$$\mathbf{s} = \dot{\mathbf{e}} + \boldsymbol{\Lambda}\mathbf{e}, \quad (4.6)$$

where

$$\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \lambda_3), \lambda_i > 0,$$

is a positive-definite diagonal matrix that determines the desired error dynamics on the sliding manifold. This transforms the original second-order trajectory-tracking problem into a first-order stabilization problem [34]. Once the system trajectories reach the manifold

$$\mathbf{s} = \mathbf{0}, \quad (4.7)$$

the error dynamics reduce to

$$\dot{\mathbf{e}} = -\mathbf{\Lambda}\mathbf{e}, \quad (4.8)$$

whose solution is given by

$$\mathbf{e}(t) = \exp(-\mathbf{\Lambda}t)\mathbf{e}(0). \quad (4.9)$$

Since $\mathbf{\Lambda}$ is positive definite, the reduced-order dynamics are exponentially stable, and the convergence rate can be explicitly adjusted through the eigenvalues of $\mathbf{\Lambda}$.

This choice provides a suitable compromise between convergence speed and control effort for the injera-dispensing task under consideration.

4.3 Equivalent Control

Differentiating the sliding variable and substituting the dynamic model of Eq. (4.1) gives

$$\dot{\mathbf{s}} = \ddot{\mathbf{e}} + \mathbf{\Lambda}\dot{\mathbf{e}} = \ddot{\mathbf{q}}_d - \ddot{\mathbf{q}} + \mathbf{\Lambda}\dot{\mathbf{e}}. \quad (4.11)$$

$$\ddot{\mathbf{q}} = \mathbf{B}^{-1}(\mathbf{q})[\boldsymbol{\tau} + \boldsymbol{\Delta} - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q})]. \quad (4.12)$$

Substituting Eq. (4.12) into Eq. (4.11) yields

$$\dot{\mathbf{s}} = \ddot{\mathbf{q}}_d + \mathbf{\Lambda}\dot{\mathbf{e}} - \mathbf{B}^{-1}(\mathbf{q})[\boldsymbol{\tau} + \boldsymbol{\Delta} - \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(\mathbf{q})]. \quad (4.13)$$

The control input is decomposed into an equivalent component and a robust switching component,

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{eq} + \boldsymbol{\tau}_{sw}, \quad (4.14)$$

where the equivalent control is chosen as

$$\boxed{\boldsymbol{\tau}_{eq} = \mathbf{B}(\mathbf{q})(\ddot{\mathbf{q}}_d + \mathbf{\Lambda}\dot{\mathbf{e}}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q})} \quad (4.15)$$

Substituting Eqs. (4.14) and (4.15) into Eq. (4.13) gives

$$\dot{\mathbf{s}} = -\mathbf{B}^{-1}(\mathbf{q})[\boldsymbol{\tau}_{sw} + \boldsymbol{\Delta}]. \quad (4.16)$$

Equation (4.16) indicates that the equivalent control compensates the nominal manipulator dynamics, leaving the closed-loop sliding dynamics affected only by the robust switching term and the lumped matched disturbance. Consequently, the control design can be reduced to synthesizing $\boldsymbol{\tau}_{sw}$ such that the sliding variable converges to a small neighborhood of the origin in finite time despite the unknown disturbance.

4.4 Adaptive Super-Twisting Switching Law

To guarantee finite-time convergence of the sliding variable while mitigating the chattering associated with conventional first-order sliding-mode control, an adaptive super-twisting algorithm (ASTA) is adopted.[35]

The classical super-twisting algorithm for a scalar sliding variable σ is given by

$$\begin{aligned} u &= -k_1 |\sigma|^{1/2} \text{sign}(\sigma) + v, \\ \dot{v} &= -k_2 \text{sign}(\sigma), \end{aligned} \quad (4.17)$$

where $k_1 > 0$ and $k_2 > 0$ are controller gains.

Unlike conventional first-order sliding mode controllers that employ discontinuous sign functions and often induce severe chattering, the present approach utilizes a smooth hyperbolic tangent function within the super-twisting framework. This modification provides a practical compromise between robustness and implementation feasibility, particularly for pneumatic and robotic actuators operating under sampling, quantization, and unmodeled dynamic effects.

For the multi-joint manipulator, the switching control law is formulated component-wise as

$$\tau_{sw} = \mathbf{B}(\mathbf{q}) \left[\mathbf{K}_1 \circ |\mathbf{s}|^{1/2} \tanh\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right) - \mathbf{v} \right], \quad (4.18)$$

where $\mathbf{K}_1 = [k_{11}, k_{12}, k_{13}]^T \in \mathbb{R}_+^3$ denotes the adaptive gain vector, $\circ$ represents the Hadamard (element-wise) product, $\mathbf{v} \in \mathbb{R}^3$ is the auxiliary super-twisting state, and $\boldsymbol{\phi} = [\phi_1, \phi_2, \phi_3]^T$ defines the boundary-layer thickness associated with each joint.

with the auxiliary dynamics

$$\dot{\mathbf{v}} = -\mathbf{K}_2 \circ \tanh\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right). \quad (4.19)$$

where $\mathbf{K}_2 \in \mathbb{R}_+^3$ denotes the second-stage super-twisting gains. To preserve the structural properties of the classical super-twisting algorithm and facilitate Lyapunov analysis, the gains are coupled through

$$\mathbf{K}_2 = \frac{\beta}{2} \mathbf{K}_1^{\circ 2}, \beta > 0, \quad (4.20)$$

where $\mathbf{K}_1^{\circ 2}$ denotes the element-wise square of $\mathbf{K}_1$. Consequently, only $\mathbf{K}_1$ is adapted online, while $\mathbf{K}_2$ is computed algebraically at each sampling instant. This coupling significantly reduces the complexity of the adaptation mechanism and preserves consistency with the strict Lyapunov framework developed for classical super-twisting algorithms.

The use of the hyperbolic tangent function in place of the discontinuous sign operator introduces a thin boundary layer around the sliding manifold, thereby suppressing chattering and reducing excessive control activity. Specifically,

$$\lim_{\phi_i \rightarrow 0} \tanh\left(\frac{s_i}{\phi_i}\right) = \text{sign}(s_i), \quad (4.21)$$

which implies that the proposed switching law asymptotically recovers the conventional super-twisting controller as the boundary-layer width approaches zero. For finite values of ϕ_i , the switching action remains smooth while retaining the strong robustness characteristics of higher-order sliding modes.

Moreover, the smooth approximation satisfies the inequality

$$\left| \tanh\left(\frac{s_i}{\phi_i}\right) - \text{sign}(s_i) \right| \leq \delta_{\phi_i}, \quad (4.22)$$

where $\delta\phi_i$ is a bounded quantity determined by the selected boundary-layer thickness. Consequently, exact finite-time convergence to the sliding manifold is replaced by practical finite-time convergence to an arbitrarily small neighborhood whose size can be reduced through appropriate tuning of ϕ_i .

The resulting adaptive super-twisting switching law therefore combines the robustness advantages of higher-order sliding modes with the implementation benefits of smooth control action, making it particularly suitable for robotic manipulation tasks requiring high precision, reduced actuator wear, and reliable operation under uncertain and time-varying disturbances.

4.5 Adaptive Gain Law

To maintain robustness under time-varying disturbances while avoiding excessive control activity near the sliding manifold, an adaptive gain mechanism is developed in which only the first-stage super-twisting gain is updated online. Unlike conventional adaptive sliding mode approaches that employ fixed boundary layers or heuristic switching thresholds, the proposed method introduces a sample-time-aware dynamic threshold that explicitly incorporates disturbance information, internal controller dynamics, and digital implementation effects. This construction enables rapid gain escalation under significant disturbances while preventing unnecessary gain growth during steady-state operation.

Because the disturbance-derivative bound L of Assumption 1 is unknown, fixed gains k_1, k_2 satisfying the sufficient conditions for global finite-time convergence [36] would have to be conservatively oversized, increasing chattering and control effort. To maintain robustness under time-varying disturbances while avoiding excessive control activity near the sliding manifold, an adaptive gain mechanism in the spirit of Plestan et al. [37] is used instead, in which only the first-stage super-twisting gain k_1 is updated online: it grows while the sliding variable remains outside a small residual band, and decays, down to a strictly positive floor, once inside that band, so that no a priori bound on L is required. Unlike conventional adaptive sliding mode approaches that employ fixed boundary layers or heuristic switching thresholds, the proposed method introduces a sample-time-aware dynamic threshold that explicitly incorporates disturbance information, internal controller dynamics, and digital implementation effects. This construction enables rapid gain escalation under significant disturbances while preventing unnecessary gain growth during steady-state operation.

A first-order differentiator is employed to estimate the temporal variation of the sliding variable. The auxiliary state $\mathbf{w} \in \mathbb{R}^3$ evolves according to

$$\dot{\mathbf{w}} = \frac{1}{\tau_f}(\mathbf{s} - \mathbf{w}), \quad (4.23)$$

where $\tau_f > 0$ denotes the differentiator time constant. The estimated sliding-surface derivative is then obtained as

$$\hat{\mathbf{s}} = \frac{1}{\tau_f}(\mathbf{s} - \mathbf{w}), \quad (4.24)$$

which yields the disturbance-activity indicator

$$\boldsymbol{\psi} = -\hat{\mathbf{s}} = -\frac{1}{\tau_f}(\mathbf{s} - \mathbf{w}). \quad (4.25)$$

The quantity $\boldsymbol{\psi}$ provides an online estimate of the local variation of the sliding dynamics and serves as a key signal in the adaptation mechanism. Large values of $|\psi_i|$ indicate rapid changes in the system behavior and consequently demand stronger switching gains, whereas small values imply that the controller is operating near the desired sliding regime.

To account for the effects of digital implementation, a dynamic threshold $\boldsymbol{\varepsilon}_t$ is introduced as

$$\boldsymbol{\varepsilon}_t = (\alpha_t + \|\boldsymbol{\psi}\| + \|\mathbf{v}(k-1)\|)T_e + \mathbf{K}_2 T_e^2, \quad (4.26)$$

where T_e denotes the sampling period, $\mathbf{v}(k-1)$ is the one-step delayed auxiliary super-twisting state, and α_t represents an adaptive scaling factor. The first term reflects the influence of the estimated disturbance dynamics and accumulated control action over one sampling interval, while the second-order term $\mathbf{K}_2 T_e^2$ accounts for the evolution of the super-twisting integral component during discrete-time implementation.

The adaptive scaling factor is defined as

$$\alpha_t = \alpha_s (\mathbf{K}_1 \circ \|\mathbf{s}\|^{1/2} + \|\boldsymbol{\psi}\|), \quad (4.27)$$

where $\alpha_s > 0$ is a user-defined adaptation coefficient. This formulation increases the adaptation rate in the presence of large tracking errors or rapidly varying disturbances while maintaining moderate gain variation during nominal operation.

The first-stage super-twisting gain is updated according to

$$\dot{k}_{1,i} = \begin{cases} \frac{\alpha_t}{\|\psi_i\| + \varepsilon_{t,i}}, & \|s_i\| > \varepsilon_{t,i}, \\ -\beta_1(k_{1,i} - k_{1,\min}), & \|s_i\| \leq \varepsilon_{t,i}, \end{cases} \quad i = 1, 2, 3, \quad (4.28)$$

where $\beta_1 > 0$ determines the decay rate toward the minimum allowable gain $k_{1,\min} > 0$. When the sliding variable exceeds the dynamic threshold, the gain increases proportionally to the estimated disturbance activity, thereby strengthening the corrective action. Conversely, once the system enters the practical sliding region, the gain smoothly converges toward a strictly positive floor, preventing excessive control effort while preserving robustness against subsequent disturbances.

To guarantee boundedness of the adaptive mechanism, the gains are constrained according to

$$k_{1,\min} \leq k_{1,i} \leq k_{1,\max}, \quad i = 1, 2, 3, \quad (4.29)$$

where $k_{1,\max}$ denotes the prescribed upper limit. Owing to the algebraic coupling introduced in [Section 4.4](#), the corresponding second-stage gains satisfy

$$k_{2,\min} = \frac{\beta}{2} k_{1,\min}^2, \quad k_{2,\max} = \frac{\beta}{2} k_{1,\max}^2, \quad (4.30)$$

which guarantees positivity and boundedness of both stages of the super-twisting algorithm without requiring an additional adaptation law for $\mathbf{K}_2$.

The proposed adaptive gain mechanism therefore integrates disturbance estimation, sample-time effects, and internal controller dynamics into a unified threshold design. Unlike conventional adaptive sliding mode strategies that employ fixed switching margins, the resulting framework automatically adjusts its sensitivity to the prevailing operating conditions, yielding improved robustness, reduced control activity, and enhanced suitability for real-time robotic manipulation tasks in uncertain, time-varying environments.

4.6 Complete Adaptive Super-Twisting Sliding Mode Controller

The proposed adaptive super-twisting sliding mode controller is obtained by combining the equivalent control component introduced in [Section 4.3](#) with the adaptive switching law presented in [Section 4.4](#). The equivalent control compensates for the nominal manipulator dynamics, whereas the switching component provides robustness against matched disturbances, parameter uncertainties, friction

variations, payload changes, and unmodeled dynamics. Consequently, the total control torque is expressed as

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{eq} + \boldsymbol{\tau}_{sw}, \quad (4.31)$$

where the equivalent control is given by

$$\boldsymbol{\tau}_{eq} = \mathbf{B}(\mathbf{q})(\ddot{\mathbf{q}}_d + \Lambda \dot{\mathbf{e}}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}), \quad (4.32)$$

and the adaptive super-twisting switching component is

$$\boldsymbol{\tau}_{sw} = \mathbf{B}(\mathbf{q}) \left[\mathbf{K}_1 \circ |\mathbf{s}|^{\frac{1}{2}} \tanh\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right) - \mathbf{v} \right]. \quad (4.33)$$

Substituting (4.32) and (4.33) into (4.31) yields the complete control law

$$\boxed{\boldsymbol{\tau} = \mathbf{B}(\mathbf{q})(\ddot{\mathbf{q}}_d + \Lambda \dot{\mathbf{e}}) + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \mathbf{B}(\mathbf{q}) \left[\mathbf{K}_1 \circ |\mathbf{s}|^{\frac{1}{2}} \tanh\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right) - \mathbf{v} \right]}. \quad (4.34)$$

The adaptive gains $\mathbf{K}_1$ are updated online according to the adaptive law presented in [Section 4.5](#), while the second-stage gains are computed algebraically as

$$\mathbf{K}_2 = \frac{\beta}{2} \mathbf{K}_1^2. \quad (4.35)$$

Consequently, only a single adaptive gain vector requires online integration, reducing computational complexity while preserving the characteristic structure of the classical super-twisting algorithm. The auxiliary state $\mathbf{v}$ is generated by integrating the super-twisting dynamics

$$\dot{\mathbf{v}} = -\mathbf{K}_2 \circ \tanh\left(\frac{\mathbf{s}}{\boldsymbol{\phi}}\right), \quad (4.36)$$

whereas the disturbance-activity estimate and adaptive threshold are obtained from the differentiator presented in [Section 4.5](#).

The resulting controller possesses a hierarchical structure consisting of three complementary components. The equivalent control compensates for the nominal nonlinear manipulator dynamics, the adaptive super-twisting term suppresses matched uncertainties while maintaining continuous control signals, and the adaptive gain mechanism automatically adjusts the switching intensity according to the **estimated disturbance** level and sampling effects. This coordinated architecture provides high tracking accuracy together with improved robustness and reduced chattering, making it particularly suitable for precision robotic manipulation tasks such as automated injera dispensing under uncertain operating conditions.

4.7 Practical Finite-Time Stability Analysis

The stability of the proposed adaptive super-twisting sliding mode controller is established using an adaptation of the strict Lyapunov framework developed for the classical super-twisting algorithm. Owing to the replacement of the discontinuous switching function by the smooth hyperbolic tangent function, the analysis establishes **practical finite-time convergence** rather than exact finite-time convergence.

Assumption 1. The desired trajectory $(\mathbf{q}_d', \dot{\mathbf{q}}_d', \ddot{\mathbf{q}}_d')$ is bounded and continuously differentiable.

Assumption 2. The lumped matched disturbance $\mathbf{d}(t)$ and its derivative satisfy

$$\|\mathbf{d}(t)\| \leq \bar{d}, \|\dot{\mathbf{d}}(t)\| \leq \bar{d}_1,$$

where $\bar{d}$ and $\bar{d}_1$ are positive constants.

Assumption 3. The adaptive gains satisfy

$$K_{1,\min} \leq K_{1,i} \leq K_{1,\max}, i = 1, 2, 3,$$

and the second-stage gains are computed as

$$K_{2,i} = \frac{\beta}{2} K_{1,i}^2,$$

thus ensuring that both adaptive gains remain positive and bounded.

Theorem 1

Under Assumptions 1–3, the proposed adaptive super-twisting sliding mode controller guarantees practical finite-time convergence of the sliding variable to a compact neighborhood of the sliding manifold. Consequently, the trajectory tracking error converges exponentially to a bounded neighborhood of the origin.

Proof

Consider the Lyapunov candidate adapted from the strict Lyapunov function of the classical super-twisting algorithm

$$V = 2\mathbf{K}_2^T \|\mathbf{s}\| + \frac{1}{2} \mathbf{v}^T \mathbf{v} + \frac{1}{2} (\mathbf{K}_1 \circ \|\mathbf{s}\|^{1/2} \circ \text{sign}(\mathbf{s}) - \mathbf{v})^T (\mathbf{K}_1 \circ \|\mathbf{s}\|^{1/2} \circ \text{sign}(\mathbf{s}) - \mathbf{v}). \quad (4.37)$$

Since $K_{1,i} > 0$ and $K_{2,i} > 0$, the Lyapunov function is positive definite and radially unbounded with respect to the augmented state $(\mathbf{s}, \mathbf{v})$.

Differentiating (4.37) along the closed-loop trajectories and substituting the proposed adaptive super-twisting controller yields

$$\dot{V} = \dot{V}_{\text{STA}} + \Delta_\phi, \quad (4.38)$$

where $\dot{V}_{\text{STA}}$ denotes the Lyapunov derivative associated with the classical super-twisting algorithm and Δ_ϕ represents the bounded approximation error introduced by replacing the discontinuous switching function with the smooth hyperbolic tangent function.

Because

$$\lim_{\phi \rightarrow 0} \tanh\left(\frac{s}{\phi}\right) = \text{sign}(s),$$

there exists a positive constant c_ϕ such that

$$\left| \tanh\left(\frac{s_i}{\phi_i}\right) - \text{sign}(s_i) \right| \leq c_\phi, \quad (4.39)$$

for all admissible operating conditions. Consequently,

$$\|\Delta_\phi\| \leq c_1\phi, \quad (4.40)$$

where $c_1 > 0$ is a bounded constant depending on the controller gains and system parameters.

Furthermore, the adaptive threshold

$$\varepsilon_t = (\alpha_t + |\psi| + |v(k-1)|)T_e + K_2T_e^2$$

maintains bounded adaptive gains while preventing excessive gain growth within the practical sliding region. Therefore, the gain adaptation law contributes no destabilizing terms to the closed-loop dynamics, and the adaptive gains remain bounded by construction.

Combining the above results gives

$$\dot{V} \leq -cV^{1/2} + c_1\phi + c_2\varepsilon_t, \quad (4.41)$$

where c , c_1 , and c_2 are positive constants.

Hence, the Lyapunov function decreases whenever

$$V^{1/2} > \frac{c_1\phi + c_2\varepsilon_t}{c},$$

implying that the sliding variable reaches, in finite time, the compact residual set

$$\Omega_s = \{\mathbf{s}: \|\mathbf{s}\| \leq C_1\phi + C_2\varepsilon_t\}, \quad (4.42)$$

where C_1 and C_2 are positive constants.

Once the sliding variable enters this residual set, the tracking error dynamics satisfy

$$\dot{\mathbf{e}} = -\mathbf{\Lambda}\mathbf{e} + \mathbf{s}. \quad (4.43)$$

Since $\mathbf{\Lambda}$ is positive definite and $\mathbf{s}$ is uniformly bounded, the tracking error converges exponentially to the neighborhood

$$\Omega_e = \left\{ \mathbf{e}: \|\mathbf{e}\| \leq \frac{C_1\phi + C_2\varepsilon_t}{\lambda_{\min}(\mathbf{\Lambda})} \right\}. \quad (4.44)$$

Therefore, the proposed adaptive super-twisting controller guarantees practical finite-time convergence of the sliding manifold together with exponential convergence of the trajectory tracking error to an arbitrarily small neighborhood of the desired trajectory by appropriate selection of the boundary-layer thickness ϕ and the adaptive threshold ε_t .

5. Simulation and Results

5.1 Simulation Setup and Controller Configurations

The proposed Adaptive Super-Twisting Sliding Mode Controller (ASTSMC) is evaluated against a fixed-gain Super-Twisting Sliding Mode Controller (STSMC) and a classical first-order Sliding Mode Controller (SMC) that employs a discontinuous switching function, under identical simulation conditions. All three controllers share the same equivalent-control component derived in Section 4.3 and are evaluated using the same nonlinear manipulator model, desired trajectories, initial conditions, and disturbance realizations. Consequently, any observed performance differences arise solely from the respective switching-law formulations.

The manipulator dynamics were simulated in MATLAB/Simulink using a fixed-step fourth-order Runge-Kutta (ode4) solver with a step size of 1×10^{-4} s, while the controllers were implemented as discrete MATLAB Function blocks operating with a sampling period of $T_e = 1 \times 10^{-3}$ s, consistent with the sample-time-aware adaptive gain law developed in Section 4.5.

The manipulator parameters correspond to a medium-sized 3-DOF SCARA robot designed for automated injera batter dispensing, having a horizontal reach of approximately 750 mm. The physical parameters used throughout all simulations are listed in Table 5.1. These parameters are employed consistently in the plant model and in the model-based equivalent control term of all three controllers.

TABLE 5.1. Manipulator Physical Parameters Used in Simulation

Parameter	Symbol	Value	Unit
Shoulder link length	l_1	0.40	m
Elbow link length	l_2	0.35	m
Fixed shoulder height	d_1	0.50	m
Link 1 mass	m_1	4.5	kg
Link 2 mass	m_2	3.2	kg
Slider (dispensing head) mass	m_3	4	kg
Link 1 inertia	I_1	0.05	kg·m ²
Link 2 inertia	I_2	0.03	kg·m ²
Slider inertia	I_3	0.001	kg·m ²
Gravitational acceleration	g	9.81	m/s ²

For all three configurations, the sliding surface was defined as $\mathbf{s} = \dot{\mathbf{e}} + \Lambda \mathbf{e}$, with the sliding surface gain matrix selected as $\Lambda = \text{diag}(5,5,5)$. While the classical SMC employed a discontinuous switching function based on the sign operator with a fixed switching gain vector $\mathbf{K}$, the STSMC replaced this discontinuous action with a continuous super-twisting algorithm using a hyperbolic tangent approximation, fixed first-stage gains $\mathbf{k}_1 = [5, 5, 5]^T$, and algebraically computed second-stage gains $\mathbf{k}_2$. The proposed ASTSMC employed the same super-twisting structure but introduced an adaptive first-stage gain updated online according to the adaptive law developed in Section 4.5. To reduce chattering while preserving robustness, the boundary-layer widths used in the switching function were selected as $\phi = [0.01, 0.01, 0.03]^T$ for the ASTSMC and $\phi = [0.01, 0.01, 0.01]^T$ for the fixed-gain STSMC. The controller configurations are summarized in Table 5.2

Table 5.2 Controller Gain Configurations

Controller	Switching Law	Controller Parameters
SMC	Discontinuous sign(s)	$\mathbf{K} = [10, 10, 10]^T$, $\Lambda = \text{diag}(5, 5, 5)$
STSMC	Super-twisting with $\tanh(s/\phi)$	$\mathbf{k}_1 = [5, 5, 5]^T$, $\beta_c = 2$, $\mathbf{k}_2 = \frac{\beta_c}{2} \mathbf{k}_1^2$, $\phi = [0.01, 0.01, 0.01]^T$
ASTSMC	Adaptive super-twisting with $\tanh(s/\phi)$	$\mathbf{k}_{\min} = [3, 3, 1.5]^T$, $\mathbf{k}_{\max} = [50, 50, 20]^T$, $\alpha_s = [8, 8, 3]^T$, $\beta_1 = [2, 2, 4]^T$, $\beta_c = 2$, $\tau_f = 0.01$ s, $\phi = [0.01, 0.01, 0.03]^T$

Gains for the prismatic axis (d_3) were tuned independently from θ_1 and θ_2 in the ASTSMC configuration, reflecting the axis's distinct dynamics: direct force actuation without rotational inertial coupling, but proportionally greater sensitivity to the disturbance and payload magnitudes considered in Section 5.5.

5.2 Nominal Tracking Performance (Step Reference)

To evaluate baseline tracking accuracy independent of disturbance and trajectory-coupling effects, the manipulator was first commanded to a single point-to-point step reference in task space. This scenario isolates each controller's intrinsic transient response, rise time, overshoot, and steady-state accuracy, independent of the continuously time-varying coupling present in the spiral trajectory considered in [Section 5.4](#).

5.2.1 Nominal Joint-Space Tracking and Error

Figure 5.1 compares the joint-space tracking responses of the three controllers for the revolute joints (θ_1 , θ_2) and the prismatic joint (d_3). All controllers successfully drive the manipulator to the desired equilibrium configuration and achieve zero steady-state tracking error under nominal conditions; however, clear differences are observed in the transient response, summarized in Table 5.3.

ASTSMC achieves the fastest rise and settling times for every joint while maintaining negligible overshoot ($<0.01\%$). STSMC settles more slowly and exhibits a pronounced overshoot of 10.1% on θ_1 and 8.0% on θ_2 , visible as the secondary oscillation in Figure 5.1(a)–(b) before both joints settle. In contrast, SMC shows no overshoot, but only because its discontinuous switching produces a slower, chattering-dominated approach rather than a clean transient as seen in Figure 5.2. These results indicate that the online adaptation of the first-stage super-twisting gain allows ASTSMC to combine fast reaching-phase convergence with low-overshoot smoothness, rather than trading one for the other as STSMC's fixed gain must. Figure 5.2 shows the corresponding tracking error, confirming that ASTSMC's error decays with visibly less oscillation than STSMC's on both revolute joints.

TABLE 5.3. Transient Response Metrics Under Step Reference

Controller	Joint	Rise Time (s)	Settling Time (s)	Overshoot (%)
SMC	θ_1	0.869	1.424	0.00
SMC	θ_2	0.697	1.210	0.00
SMC	d_3	0.443	0.827	0.00
STSMC	θ_1	0.542	1.458	10.10
STSMC	θ_2	0.484	1.243	8.03
STSMC	d_3	0.403	0.812	0.00
ASTSMC	θ_1	0.466	0.947	0.00
ASTSMC	θ_2	0.460	0.927	0.00
ASTSMC	d_3	0.426	0.833	0.01

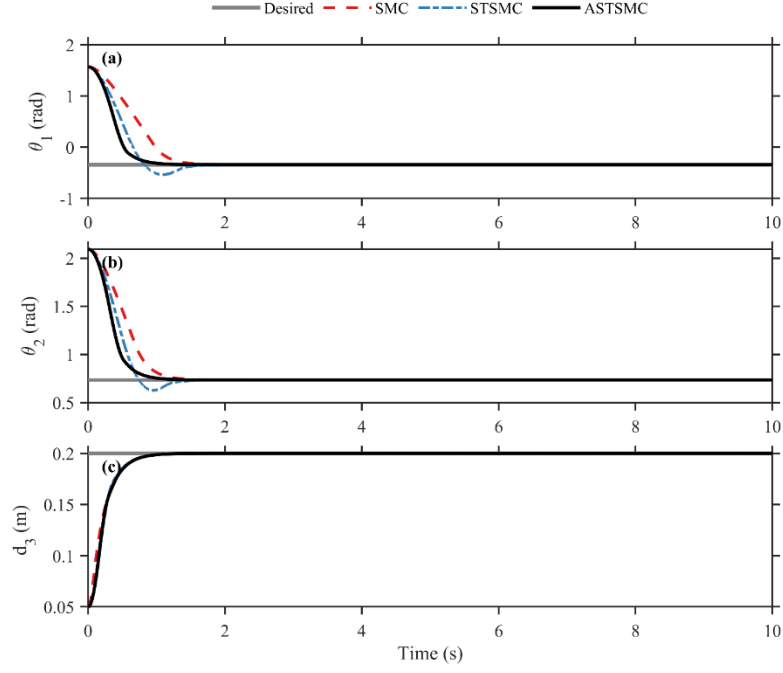


Fig. 5.1. Fig. 5.1. Joint-space tracking response under step reference: (a) θ_1 , (b) θ_2 , (c) d_3 .

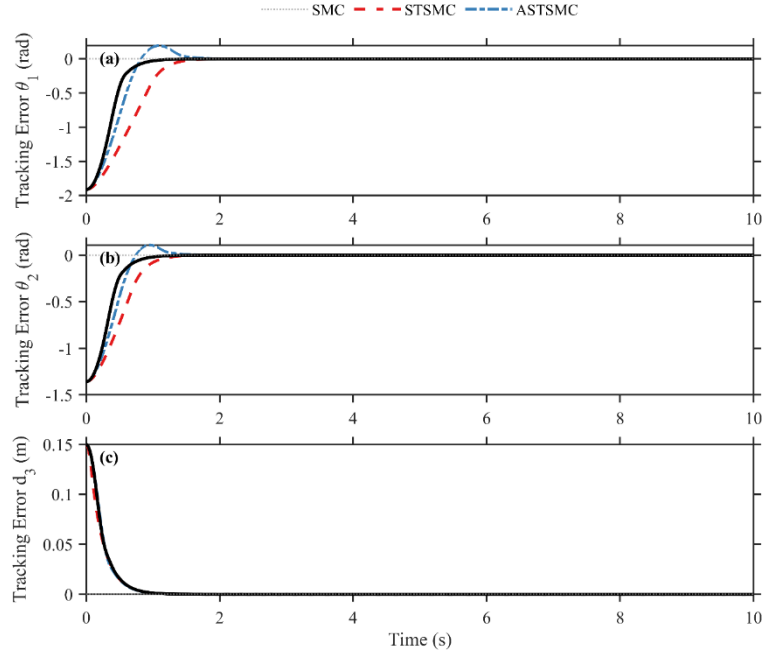


Fig. 5.2. Joint-space tracking error under step reference for all three controllers.

5.2.2 Sliding Surface and Control Signal

Figure 5.3 illustrates the evolution of the sliding variables $s(t)$ for all three joints. For each controller, the sliding variables converge rapidly toward the vicinity of the origin, confirming successful establishment of the sliding motion; the classical SMC exhibits comparatively larger oscillations around the sliding manifold due to its discontinuous switching action, whereas both super-twisting controllers generate significantly smoother sliding trajectories. Among the three, ASTSMC reaches the practical sliding region most rapidly while maintaining the smallest residual oscillations, consistent with the practical finite-time stability analysis of Section 4.7, where the adaptive gain mechanism enables rapid convergence without requiring an excessively conservative fixed gain.

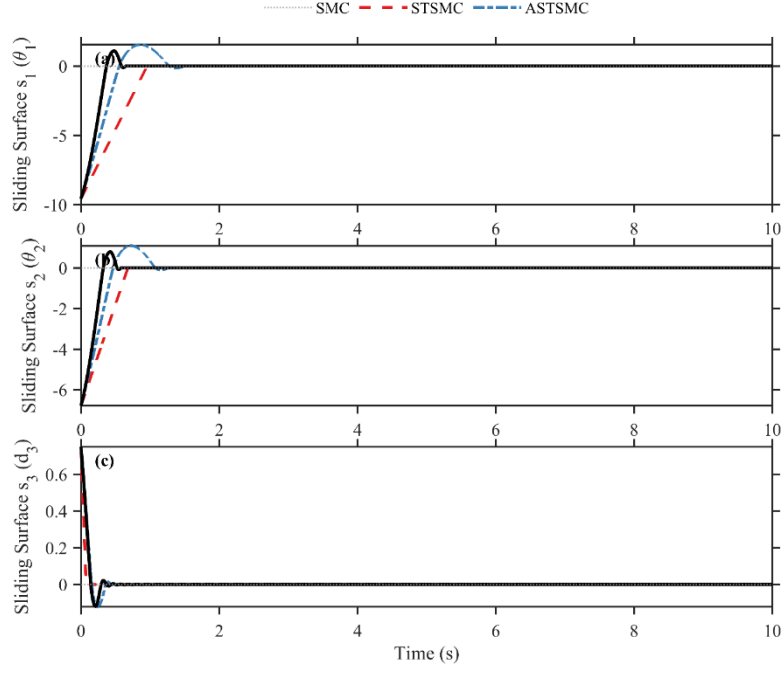


Fig. 5.3. Sliding surface $s(t)$ under step reference for all three controllers.

This behavior is directly reflected in the corresponding actuator torque and force commands, shown in Figure 5.4: the discontinuous switching law of the classical SMC produces the largest control activity and the highest chattering level, the super-twisting algorithm significantly smooths the control signals by replacing this discontinuous action with a continuous control law, and the proposed ASTSMC maintains control signals comparable in smoothness to those of the fixed-gain STSMC while automatically adjusting the switching gains according to the tracking condition, consequently improving tracking performance without introducing excessive actuator demand.

Quantitative evaluation further confirms the superiority of the proposed controller. The total ITAE values obtained for the SMC, STSMC, and ASTSMC are 0.81654, 0.51903, and 0.30800, respectively. **Relative to the classical SMC, the proposed ASTSMC reduces the total ITAE by approximately 62.3%; relative to the fixed-gain STSMC, the reduction is 40.7%.**

The corresponding RMS control efforts are 59.657, 40.129, and 41.287, respectively. Thus, compared with the STSMC, the proposed controller achieves substantially higher tracking accuracy with only a 2.9% increase in RMS control effort, demonstrating that the adaptive gain mechanism significantly improves tracking precision while requiring only a negligible increase in actuator effort under nominal operating conditions. A quantitative comparison of the tracking performance and control effort for all controllers is provided in **Table 5.3.**

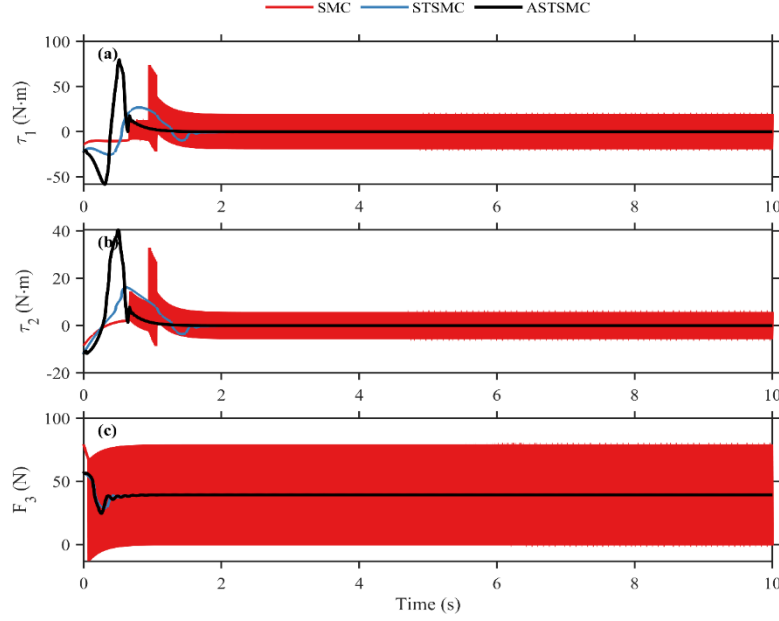


Fig. 5.4. Control signals τ_1 , τ_2 , F_3 under step reference for all three controllers.

5.3 Spiral Trajectory Tracking

The manipulator was next commanded to follow the Archimedean inward spiral trajectory of Eq. (2.27)–(2.29), representative of the intended injera-dispensing motion, with the disturbance generator set to Mode 1 (None). Unlike the step case of Section 5.3, this scenario exercises the fully coupled, θ_2 -dependent inertia and Coriolis terms of Eqs. (3.6)–(3.8) continuously throughout the motion, providing a more demanding assessment of tracking accuracy under time-varying dynamics.

5.3.1 Joint-Space Tracking and Error

Figure 5.5 illustrates the joint-space tracking responses of the three controllers during execution of the spiral trajectory. All controllers successfully follow the desired joint trajectories throughout the motion; however, noticeable differences exist in tracking accuracy. The classical SMC exhibits the largest tracking deviations, particularly during periods of rapidly changing trajectory curvature where the nonlinear coupling between the revolute joints becomes more pronounced. The STSMC significantly improves the tracking performance by reducing transient oscillations through the continuous super-twisting algorithm. The proposed ASTSMC provides the closest agreement between the desired and actual joint trajectories over the entire motion, demonstrating that the adaptive gain mechanism effectively compensates for the continuously varying nonlinear dynamics while maintaining smooth convergence

Figure 5.6 presents the corresponding joint-space tracking error for each controller, complementing the overlay comparison of Figure 5.5 by isolating the magnitude of deviation that is otherwise difficult to discern when the actual and desired trajectories lie close together. Consistent with the qualitative trends observed above, the per-joint ITAE values confirm that the shoulder joint (θ_1) contributes the largest share of tracking error for all three controllers (0.5575, 0.3360, and 0.1834 for SMC, STSMC, and ASTSMC, respectively), reflecting its greater exposure to the coupled inertial and Coriolis terms of Eq. (3.6). The elbow joint (θ_2) contributes a comparatively smaller but still substantial error component, while the prismatic axis (d_3) records the smallest error across all controllers (0.0201, 0.0076, and 0.0085, respectively), consistent with its decoupled, gravity-dominated dynamics (Eq. 3.38) that are largely unaffected by the continuously varying coupling terms driving θ_1 and θ_2 . The proposed ASTSMC achieves the lowest error on both revolute joints, reducing θ_1 error by 45.4% and θ_2 error by 33.5% relative to STSMC, indicating that the adaptive gain mechanism is particularly effective in compensating for the trajectory-coupled dynamics responsible for the largest error contributions.

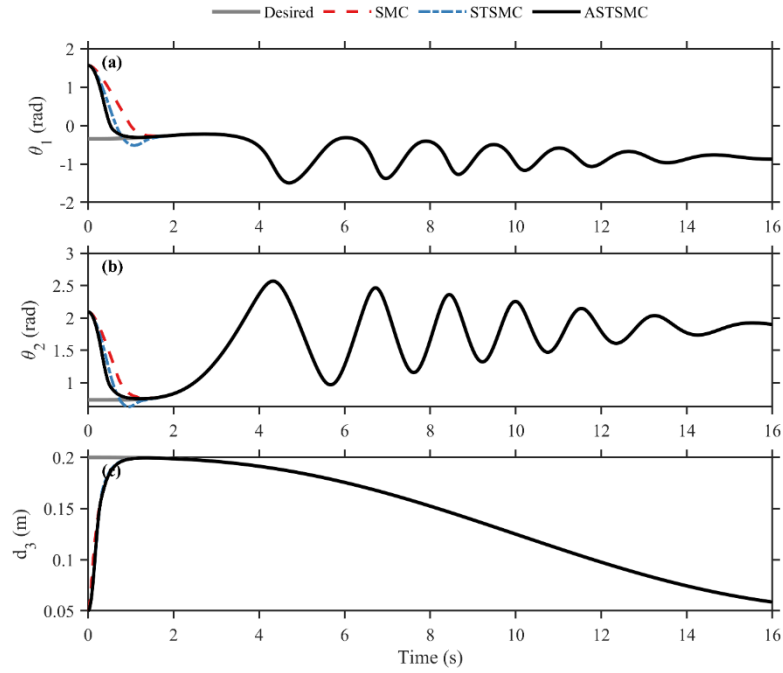


Fig. 5.5. Joint-space tracking response during spiral trajectory execution.

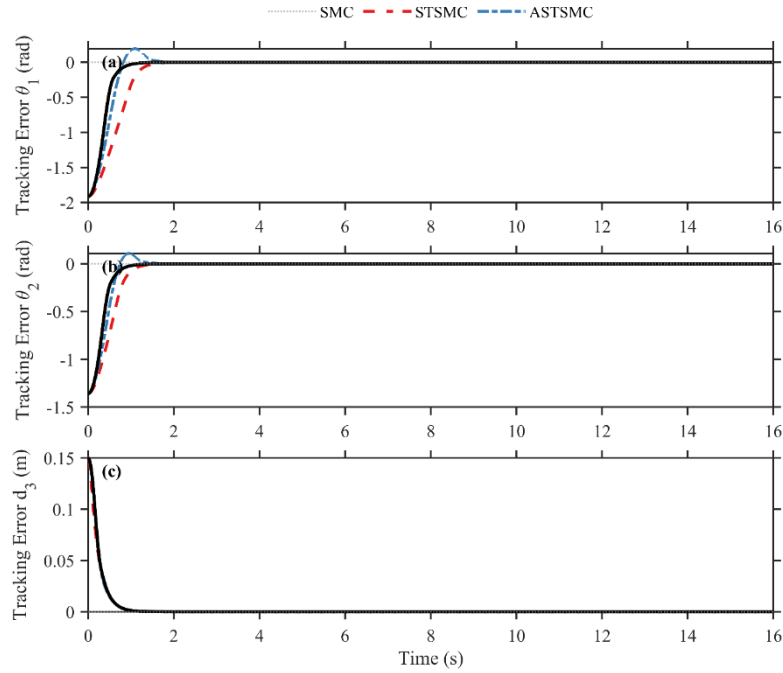


Fig. 5.6. Joint-space tracking error during nominal spiral trajectory execution.

5.3.2 Task-Space (Cartesian) Tracking

Figure 5.7 presents the Cartesian tracking performance of the end-effector. Since the dispensing quality depends directly on the nozzle position rather than individual joint variables, Cartesian tracking accuracy provides a practical measure of controller performance. All three controllers maintain stable trajectory tracking; however, the ASTSMC produces the smallest deviation between the desired and actual end-effector position in the X-Y plane throughout the spiral motion. This improvement is particularly evident during regions of high trajectory curvature, where accurate coordination of the two revolute joints is essential for maintaining the desired dispensing path, consistent with the θ_1 and θ_2

tracking-error reductions of 45.4% and 33.5% reported in Section 5.4.1, since X and Y are direct trigonometric functions of these two joint angles. The prismatic axis (d_3), governing the vertical dispensing height, follows the trend noted in Section 5.4.1 and is discussed there rather than repeated here.

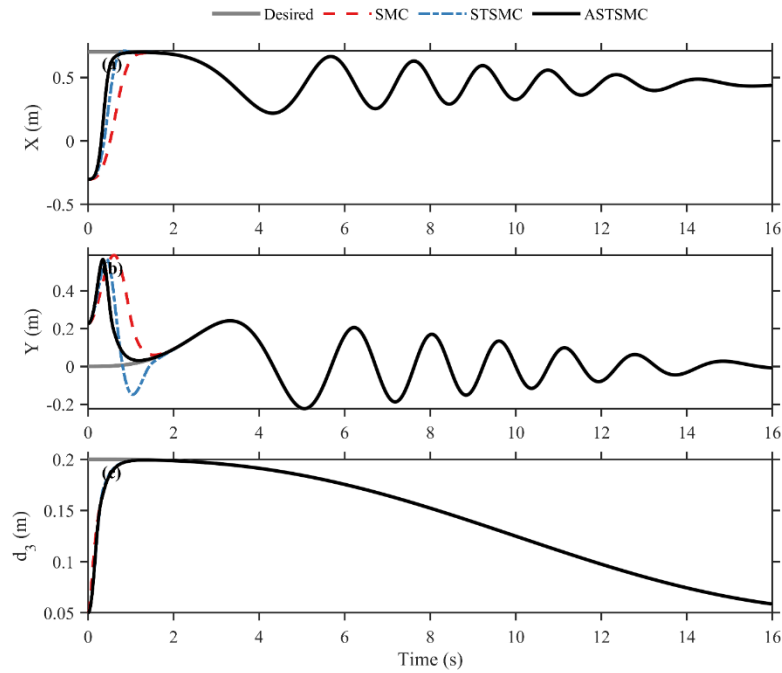


Fig. 5.7. Task-space (Cartesian) tracking response during spiral trajectory execution.

5.3.3 XY Spiral Path Tracking

Figure 5.8 compares the desired and actual end-effector trajectories in the XY-plane. The desired Archimedean spiral is accurately reproduced by all controllers, confirming successful execution of the planned dispensing path. Nevertheless, the classical SMC exhibits visibly larger path deviations, whereas the STSMC produces a considerably smoother trajectory. The proposed ASTSMC achieves the closest agreement with the desired spiral, yielding the smallest path deviation over the entire dispensing operation. From a practical perspective, improved path fidelity directly contributes to more uniform batter deposition and consequently enhances the geometric consistency and thickness distribution of the produced injera

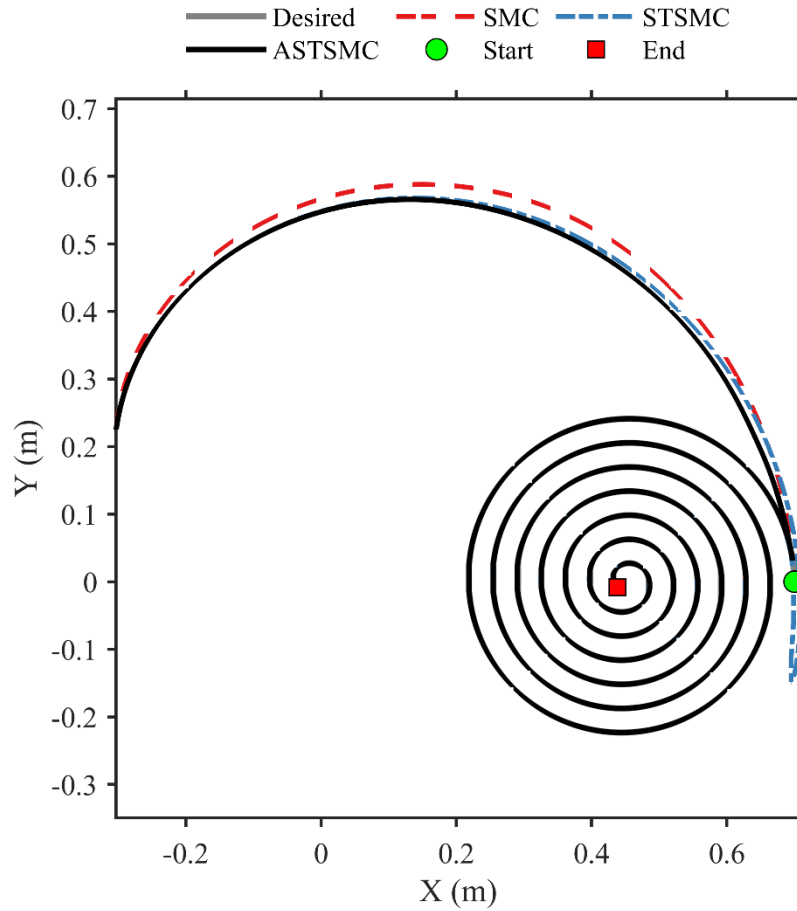


Fig. 5.8. Actual vs. desired end-effector path in the XY-plane for the injera-dispensing spiral trajectory.

5.3.4 Control Signal

Figure 5.9 compares the actuator torque and force commands during spiral trajectory execution. Owing to the continuously varying manipulator dynamics, all controllers require larger control activity than in the step-tracking experiment. Nevertheless, the super-twisting controllers generate substantially smoother control signals than the classical SMC. The proposed ASTSMC automatically adjusts its switching gain according to the instantaneous tracking condition, thereby maintaining high tracking accuracy without excessive control effort.

Quantitatively, the total ITAE values obtained for the SMC, STSMC, and ASTSMC are 0.83460, 0.51906, and 0.30844, respectively. Relative to the classical SMC, the proposed ASTSMC reduces the total ITAE by approximately 63.0%; relative to the fixed-gain STSMC, the reduction is 40.6%. The corresponding RMS control efforts are 59.726, 39.998, and 40.728, respectively. Compared with the fixed-gain STSMC, the proposed controller therefore achieves substantially higher tracking accuracy with only a 1.8% increase in RMS control effort, demonstrating that adaptive gain scheduling effectively improves trajectory tracking under continuously varying nonlinear dynamics. The corresponding quantitative performance metrics are summarized in Table 5.3 together with those obtained for the remaining experimental scenarios.

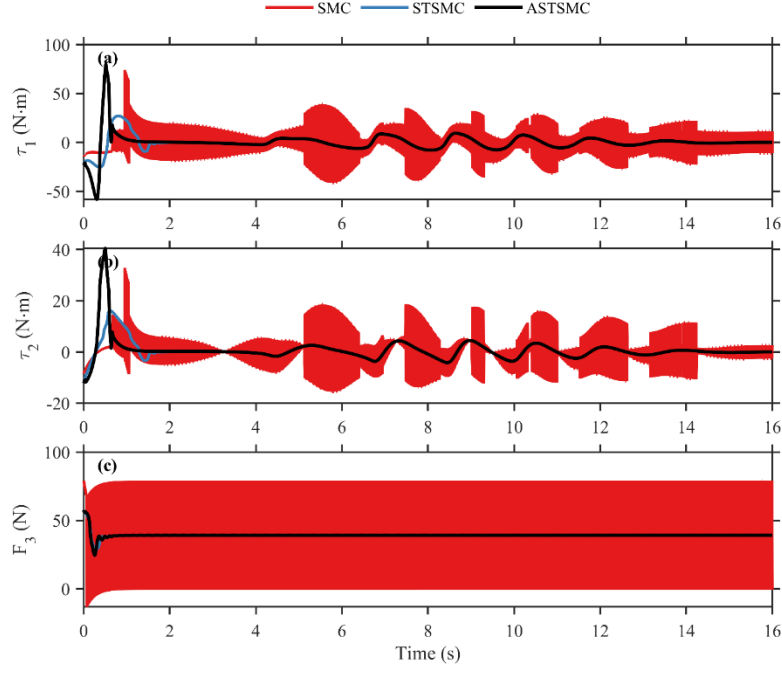


Fig. 5.9. Control signals τ_1 , τ_2 , F_3 during nominal spiral trajectory execution.

5.5 Robustness Analysis

To evaluate robustness under the conditions motivating the adaptive gain law of Section 4.5, two further scenarios were considered: (i) an external matched disturbance introduced during the spiral trajectory, and (ii) a mismatch between the physical parameters used in the plant and those assumed by the controller's model-based equivalent-control term, representing parametric uncertainty.

5.5.1 Disturbance Rejection Under Point-to-Point Motion

The step reference of Section 5.3 was repeated with a smoothed step disturbance (Eq. 4.2) introduced at $t = 5$ s. ASTSMC achieved a total ITAE of 0.410, a 34.8% improvement relative to STSMC's 0.628, with an RMS control effort of 42.31 versus STSMC's 41.17 (2.8% higher). Relative to SMC (ITAE 0.819, RMS effort 63.42), ASTSMC reduced tracking error by 50.0% and control effort by 33.3%.

Figure 5.10 shows the joint-space tracking response. All controllers converge during the initial reaching phase; the disturbance onset at $t = 5$ s then appears as a secondary transient, smallest and most quickly attenuated for ASTSMC. Figure 5.11 shows the corresponding tracking error, isolating this second excursion more clearly: largest and slowest to recover for SMC, intermediate for STSMC, and smallest for ASTSMC.

Figure 5.12 shows the sliding surface $s(t)$, which returns to a small neighborhood of the origin after the reaching phase (consistent with Eq. 4.16) and shows a brief, similarly ranked departure at disturbance onset. Figure 5.13 presents the control signals τ_1 , τ_2 , and F_3 ; SMC's discontinuous switching produces visibly higher chattering at $t = 5$ s than either super-twisting variant, while ASTSMC and STSMC remain comparably smooth.

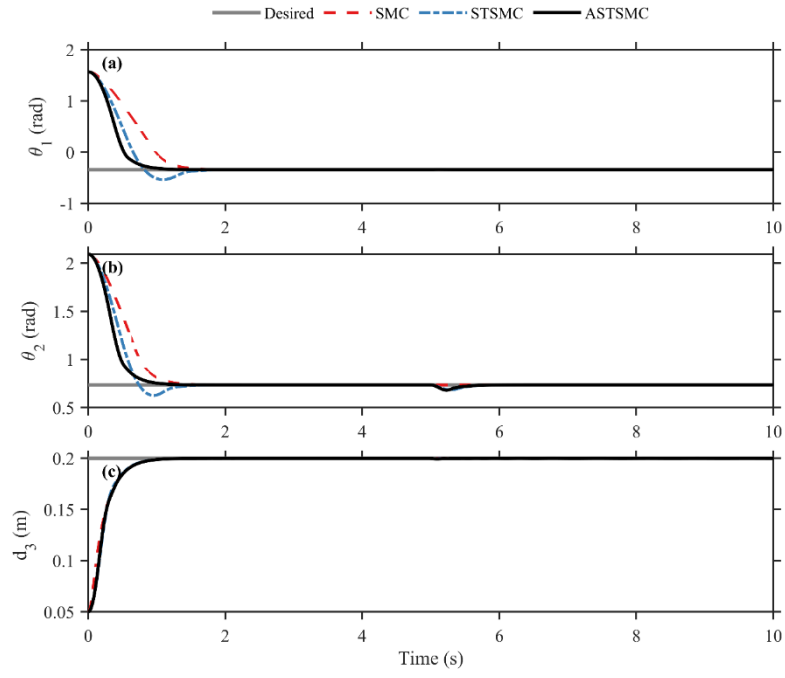


Fig. 5.10. Joint-space tracking response under step reference with disturbance onset at $t = 5$ s.

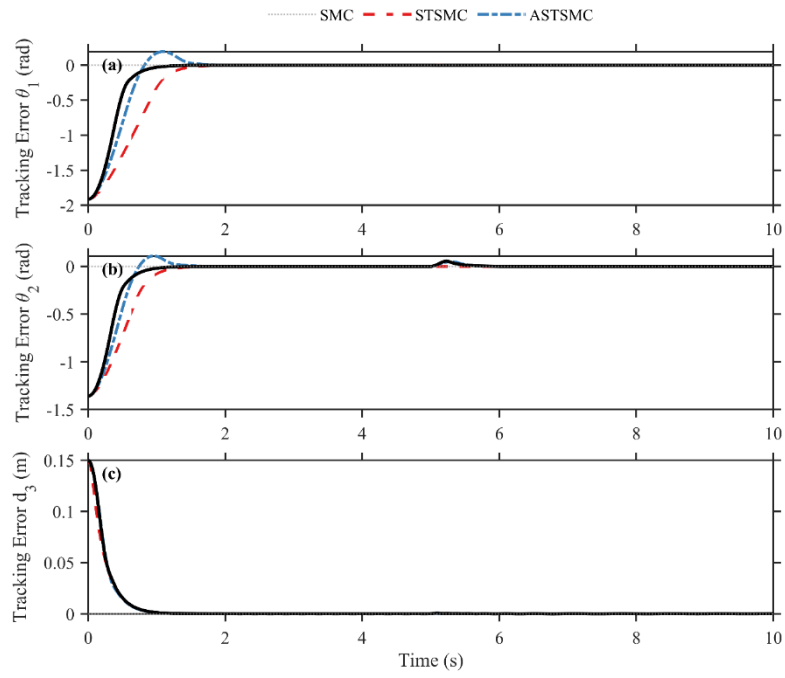


Fig. 5.11. Joint-space tracking error under step reference with disturbance onset at $t = 5$ s.

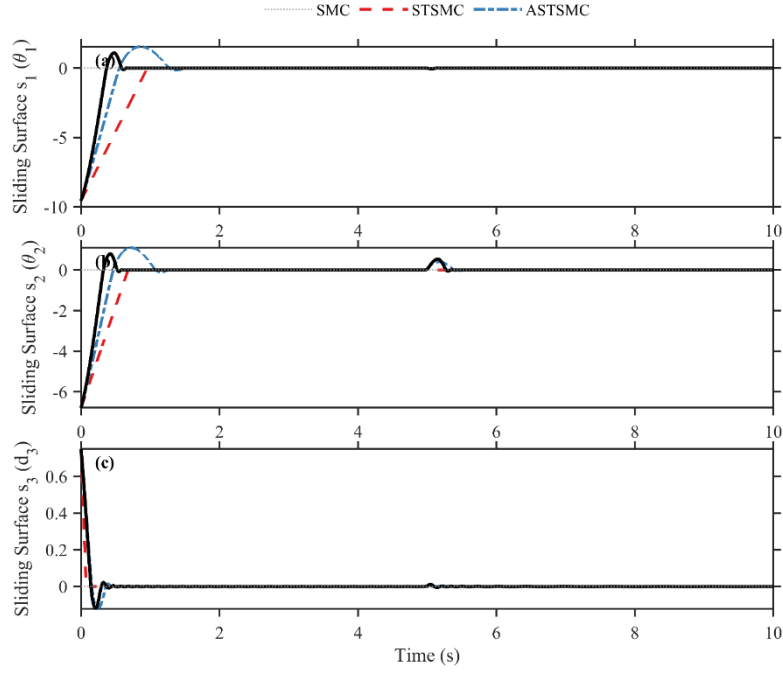


Fig. 5.12. Sliding surface $s(t)$ under step reference with disturbance onset at $t = 5$ s.

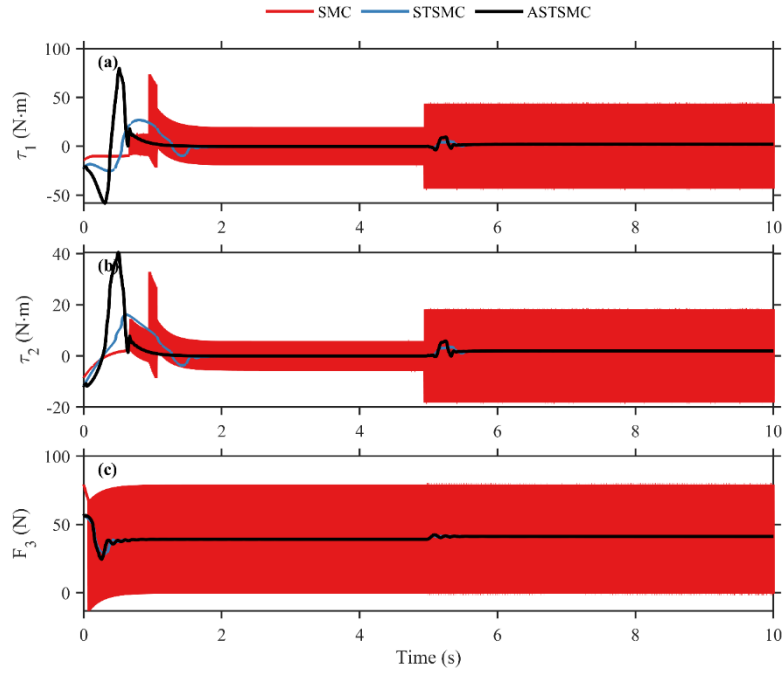


Fig. 5.13. Control signals τ_1 , τ_2 , F_3 under step reference with disturbance onset at $t = 5$ s.

Figure 5.14 shows the disturbance signal $\Delta(t)$ alongside the corresponding evolution of the ASTSMC's adaptive gain $k_1(t)$. [DISCUSSION: discuss whether k_1 rises promptly upon disturbance onset at $t = 5$ s and relaxes toward $k_{1,min}$ once the tracking error returns to the practical sliding neighborhood, consistent with the two-branch adaptive law of Eq. (4.28).]

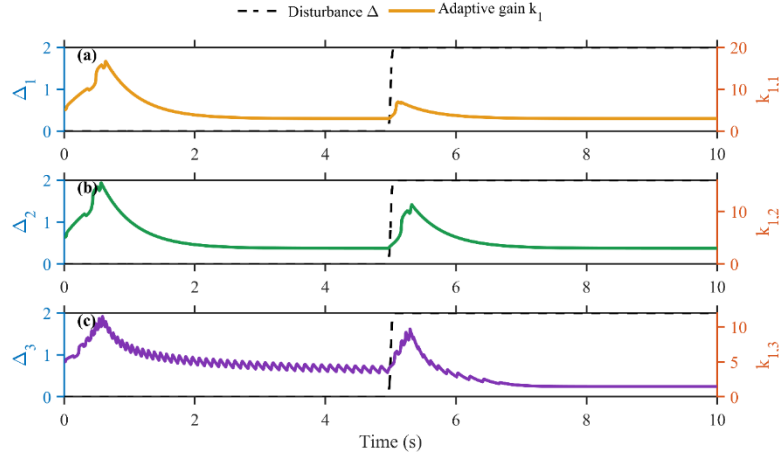


Fig. 5.14. Disturbance signal $\Delta(t)$ and corresponding adaptive gain $k_i(t)$ evolution (ASTSMC), disturbance onset at $t = 5$ s, point-to-point motion.

5.5.2 Disturbance Rejection Under Continuous Motion

The nominal spiral trajectory of [Section 5.4](#) was repeated with a matched step disturbance (Eq. 4.2) introduced at $t = 5$ s. ASTSMC achieved a total ITAE of 0.356, a 34.5% improvement relative to STSMC's 0.543, with an RMS control effort of 42.13 versus STSMC's 41.43 (1.7% higher). Relative to SMC (ITAE 0.821, RMS effort 60.25), ASTSMC reduced tracking error by 56.6% and control effort by 30.1%.

Figure 5.15 shows the joint-space tracking response; all three controllers preserve closed-loop stability through the disturbance, but SMC exhibits the largest transient deviation at onset, with STSMC improving on this and ASTSMC showing the smallest deviation and fastest recovery. Figure 5.16 shows the corresponding tracking error, isolating this disturbance-induced excursion more clearly across the same controller ranking.

Figure 5.17 shows the sliding surface $s(t)$, which, consistent with Eq. (4.16), briefly departs from the origin at disturbance onset before reconverging under the switching term. Figure 5.18 shows the control signals τ_1 , τ_2 , F_3 ; SMC's discontinuous switching produces visibly more chattering at $t = 5$ s than either super-twisting variant, while ASTSMC and STSMC remain comparably smooth despite ASTSMC's marginally higher effort.

Figure 5.19 shows the disturbance $\Delta(t)$ alongside ASTSMC's adaptive gain $k_i(t)$. Following disturbance onset, k_i rises automatically in response to the elevated sliding variable, strengthening the switching action during the recovery transient, then relaxes back toward $k_{i,\min}$ once the tracking error re-enters the practical sliding region — consistent with the adaptive law of Eq. (4.28), and demonstrating that the controller supplies additional robustness only when required rather than sustaining a conservatively large fixed gain throughout, as STSMC's constant gain must.

This result is consistent with both the point-to-point disturbance case of Section 5.5.1 and the two nominal scenarios of Sections 5.3–5.4: ASTSMC's accuracy advantage over STSMC (34.5–40.7% across all four scenarios) holds regardless of reference type or disturbance presence, at a consistently small effort cost (1.7–2.9%). One nuance worth noting: ASTSMC's own ITAE degrades proportionally more under disturbance than STSMC's (33.1% vs. 21.0% for step; 15.5% vs. 4.7% for spiral, each relative to its own undisturbed baseline), yet its absolute ITAE remains lower than STSMC's in every case, since it starts from a substantially lower undisturbed baseline.

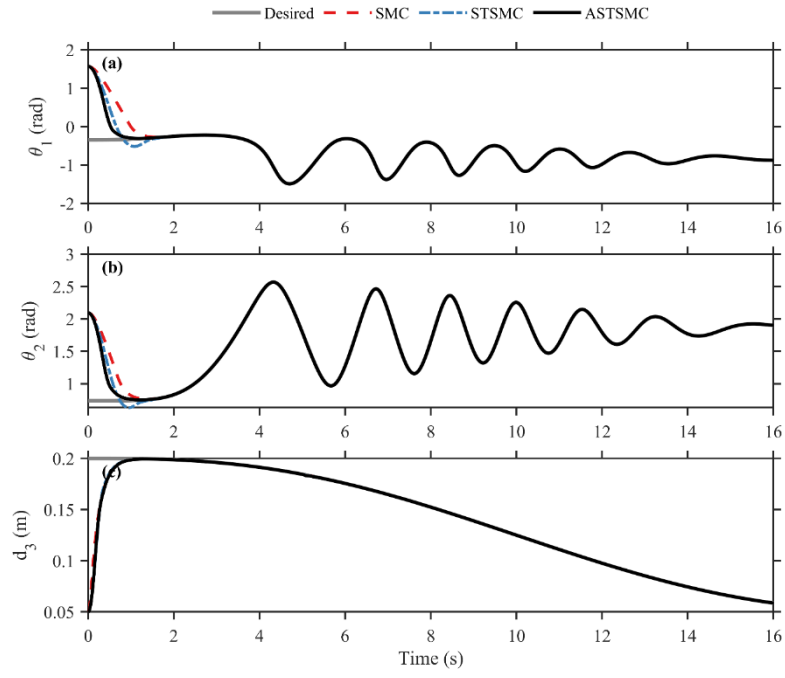


Fig. 5.15. Joint-space tracking response during spiral trajectory with disturbance onset at $t = 5$ s.

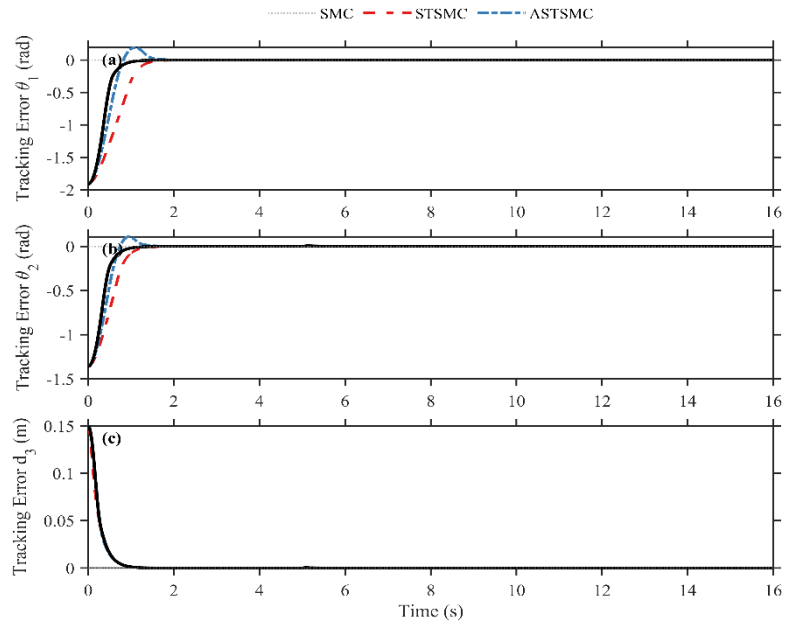


Fig. 5.16. Joint tracking error during spiral trajectory with disturbance onset at $t = 5$ s.

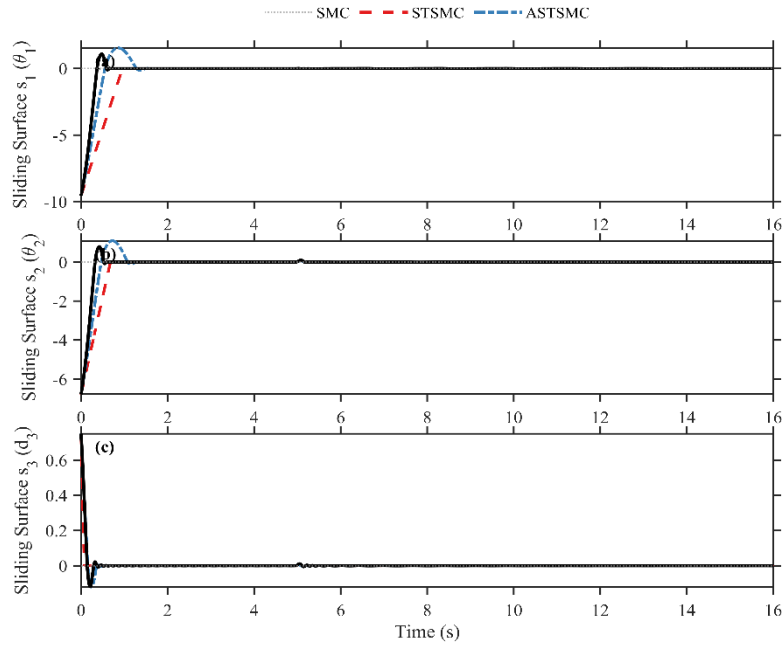


Fig. 5.17. Sliding surface $s(t)$ during spiral trajectory with disturbance onset at $t = 5$ s.

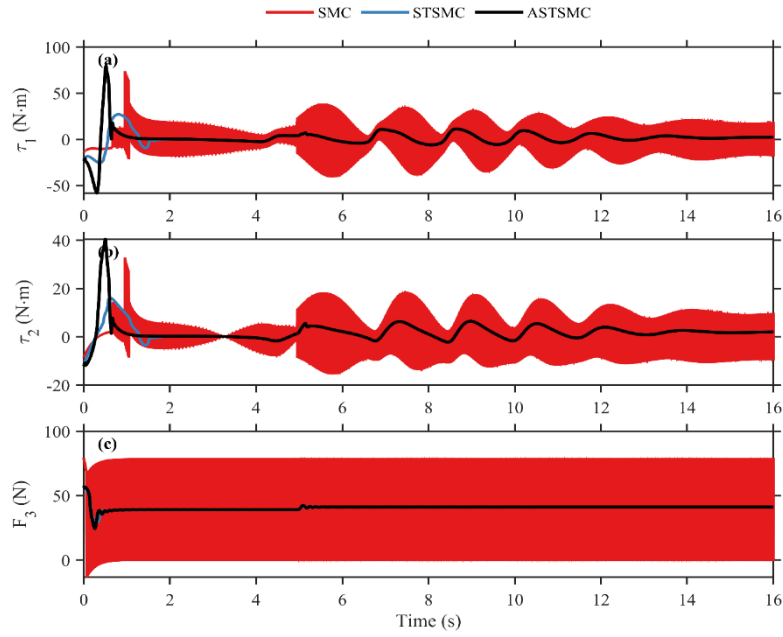


Fig. 5.18. Control signals τ_1 , τ_2 , F_3 during spiral trajectory with disturbance onset at $t = 5$ s.

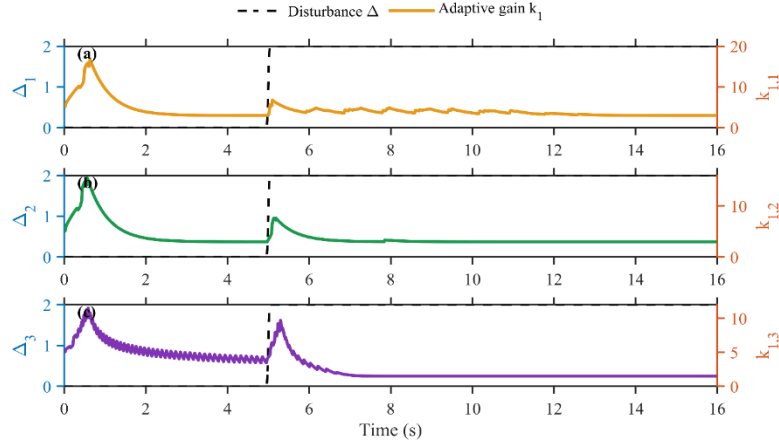


Fig. 5.19. Disturbance signal $\Delta(t)$ and corresponding adaptive gain $k_i(t)$ evolution (ASTSMC), disturbance onset at $t = 5$ s, continuous spiral motion.

5.5.2 Robustness Against Parameter Uncertainty

To evaluate robustness against modeling uncertainty, a 20% payload mass perturbation was introduced by increasing the prismatic-link mass from 4.0 kg to 4.8 kg in the plant model, while all controllers retained the nominal parameters. No external disturbance was applied, allowing the effects of parameter mismatch to be evaluated independently. The Archimedean spiral trajectory was used because it represents the intended injera-dispensing operation.

Figure 5.20 compares the joint tracking errors under the perturbed model. All controllers maintained closed-loop stability; however, the proposed ASTSMC exhibited the smallest tracking errors owing to its adaptive gain mechanism, which automatically compensated for the increased model uncertainty.

Quantitatively, the SMC, STSMC, and ASTSMC achieved total ITAE values of 0.89808, 0.77479, and 0.32640, respectively. Compared with the SMC and STSMC, the proposed controller reduced the total ITAE by 63.7% and 57.9%, respectively. The corresponding RMS control efforts were 64.911, 47.936, and 48.980, indicating that the improved tracking performance was achieved with only a 2.2% increase in control effort relative to the STSMC. These results demonstrate the robustness of the proposed ASTSMC against moderate payload variations and model uncertainty.

The corresponding quantitative performance metrics are summarized in Table 5.3.

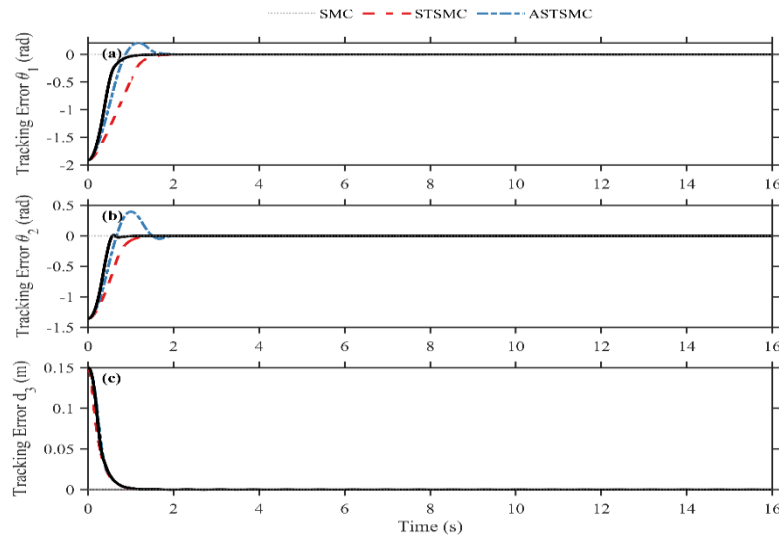


Fig. 5.20. Joint tracking error under a 20% payload mass perturbation ($m_3 = 4.8$ kg).

5.6 Overall Quantitative Performance Comparison

Table 5.3 summarizes the Integral of Time-weighted Absolute Error (ITAE) and control-effort norms (RMS torque/force) for all three controllers across the five test scenarios of Sections 5.3–5.5. ITAE was computed as $\int_0^T t \cdot |e(t)| dt$, penalizing persistent late-time error more heavily than transient error; the RMS control-effort norm was computed as the time-averaged L2 norm of the combined three-joint control vector.

TABLE 5.3. ITAE and Control-Effort Comparison Across Test Scenarios

Scenario	Controller	ITAE	τ_{RMS}
Step	SMC	0.81654	59.657
	STSMC	0.51903	40.129
	ASTSMC	0.30800	41.287
Step + Disturbance	SMC	0.81885	63.416
	STSMC	0.62787	41.172
	ASTSMC	0.40960	42.309
Spiral	SMC	0.83460	59.726
	STSMC	0.51906	39.998
	ASTSMC	0.30844	40.728
Spiral + Disturbance	SMC	0.82102	60.252
	STSMC	0.54349	41.428
	ASTSMC	0.35601	42.130
20% Payload Mass Variation	SMC	0.89808	64.911
	STSMC	0.77479	47.936
	ASTSMC	0.32640	48.980

The quantitative results consistently demonstrate the superiority of the proposed adaptive super-twisting sliding mode controller across all evaluated operating conditions. Under nominal step tracking, the ASTSMC achieves the lowest tracking error, reducing the total ITAE by 62.3% relative to the classical SMC and by 40.7% relative to the fixed-gain STSMC. Similar improvements are observed during the spiral trajectory, where the proposed controller reduces the total ITAE by 63.1% and 40.6%, respectively.

Under disturbed operating conditions, the adaptive gain mechanism maintains its effectiveness. For the step-disturbance experiment, the ASTSMC decreases the total ITAE by 50.0% compared with the SMC and by 34.8% compared with the STSMC. Likewise, during the spiral trajectory with an external disturbance, the proposed controller achieves ITAE reductions of 56.6% and 34.5%, respectively.

The robustness of the proposed controller is further demonstrated under a 20% payload mass variation, where ASTSMC achieves a total ITAE reduction of 63.7% relative to the classical SMC and 57.9% relative to the fixed-gain STSMC. Although the RMS control effort increases by only 2.2% compared with the STSMC, the substantial improvement in tracking accuracy confirms the effectiveness of the adaptive gain mechanism in compensating for model uncertainty.

Although the adaptive gain mechanism introduces a slight increase in actuator effort compared with the fixed-gain STSMC, this increase remains modest, ranging from approximately 1.7% to 2.9% across all test scenarios. In contrast, the classical SMC consistently requires substantially greater control effort while producing the largest tracking errors. These results demonstrate that the proposed ASTSMC provides a favorable compromise between tracking accuracy and actuator demand, yielding significantly improved trajectory tracking with only a negligible increase in control effort relative to the fixed-gain super-twisting controller.

6. Conclusion

An Adaptive Super-Twisting Sliding Mode Controller (ASTSMC) was proposed in this research work for robust trajectory tracking of a 3-DOF SCARA robot designed for the automated injera batter dispensing. This ASTSMC design incorporates a smooth switching function represented by a hyperbolic tangent together with an adaptive gain method to enhance tracking accuracy while reducing the chattering inherent in sliding mode control systems. Practical finite-time stability was established using a Lyapunov-based analysis, demonstrating convergence of the tracking errors to a bounded neighborhood of the sliding manifold.

The proposed controller was evaluated using five simulation scenarios, which include nominal step tracking, step tracking with disturbance, nominal spiral tracking, spiral tracking with disturbance, and a 20% change in payload mass representing model uncertainty. In every case, ASTSMC consistently outperformed both the classical SMC and the fixed-gain STSMC. Relative to the STSMC, the proposed controller reduced the total ITAE by 40.7% for nominal step tracking, 34.8% under step disturbance, 40.6% during nominal spiral tracking, 34.5% during disturbed spiral tracking, and 57.9% under payload mass variation, while requiring only a modest increase in RMS control effort of approximately 1.7–2.9%. Compared with the classical SMC, ASTSMC reduced the total ITAE by 50.0–63.7% and simultaneously lowered the overall control effort.

Simulation results reveal that the adaptive gain approach automatically increases the switching gain in the presence of any disturbance or modeling uncertainties and decreases automatically when the tracking error gets closer to the practical sliding manifold. This mechanism enhances robustness against external perturbations and parameter variations without requiring excessively conservative, high-gain values. Furthermore, the precise tracking of the desired inward Archimedean spiral trajectory validates the feasibility of this control scheme for automated injera batter dispensing.

While these findings are currently simulation-based, the ASTSMC represents a highly promising control strategy for nonlinear manipulators operating in uncertain environments. The future research will focus on the experimental implementation of the developed approach, the investigation of the different payload cases and unmodelled dynamics, as well as the extension of the proposed adaptive super-twisting scheme to vision-based and learning robotic manipulations.

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